# Threshold Entry in a Fixed-Prize Contest

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Manuscript skeleton, 7 October 2026

## 1 Introduction

The question is how differences in wealth shape the competition for a position awarded by rank, that is, how many compete, who they are, and how much of the prize they take. The model keeps apart the two things Weber [1978] separates as class and status. Wealth sets only the cost of entering, and the prize is won by rank (rows L14 to L16 give the passages and where the parallel ends).

Table 1: Headline claims, each traced to the model rows it rests on.

| ID | Headline | Model rows | Scope note |
|---|---|---|---|
| K1 | Every pure-strategy equilibrium has the same number of entrants, $k^*$, and the $k^*$ richest challengers entering is always one of them. | M5, M6, M8, M30 | The count is pinned, and who enters is not, unless the condition of M7 holds. The count needs a gain that depends on the number of entrants alone (M26), and for some costs it fails whenever the ability laws differ by identity, falling with wealth (M27) or rising with it (M29). An incumbent who competes under the same rule is the lowest-cost player and sits out while a challenger enters only inside the band of M6 (M30). |
| K2 | The effect of a spread of wealth on how many enter has a sign set by one comparison. Entry weakly rises when the marginal entrant stands at or above the pivot, and weakly falls when it stands at or below. | M12, M13, M15 | Proved for pivot-spreads, and more generally under the margin condition. Inside the band the signs of the change settle nothing (M14), and the mean cannot replace the pivot (M13). Strict change is shown only for linear spreads (M15). |
| K3 | The sorting needs only that the fee's utility cost falls with wealth, which strict concavity implies and which is strictly weaker. | M23, M24 | No condition on $u'''$ is used. Wealth sorting into a tournament is in print (Lazear and Rosen), through a condition on $u'''$, so the mechanism is not claimed as new. |
| K4 | Entry is capped by the prize for every number of challengers. An entrant at index $m$ with cost $\kappa$ has $(m+1)\kappa \le V$. | M10 | The accounting behind the cap is in print for identical agents (Fu, Jiao and Lu; Fu and Lu). It replaces the old mobility ceiling and is not claimed as new. |
| K5 | A larger prize does not lower the number of entrants, and a larger fee does not raise it. | M11 | Weak monotonicity only. |
| K6 | When the score gap is single-peaked, the first entrant's gain can rise and then fall in the number of challengers, and it never falls and then rises. With a uniform base score it never falls, for some base scores it must eventually fall, and stochastic dominance alone does not sign it. | M16, M17, M18, M19, M20, M21, M22 | The absence of an interior minimum is Ryvkin and Drugov's Karlin step applied to the gain, with the score gap in place of the noise density, and the message that no such prediction is universal is theirs (C14). What may be new is the uniform case and the conditions on the base score (C7, pending). |
| K7 | Those who pay take the prize with a probability that does not fall as more of them pay. It equals $\frac{k^*+1}{Q+1}$ when paying buys nothing, and it does not rise, nor does the number who pay, as the rule weighs talent more. | M28, M31 | The share counts positions and not their pay; earnings after entry are outside the model (Appendix D). The directions in the weight on talent hold for every law of talent and of the bought component, with the incumbent scored by the same rule. Whether either is new is pending (C12, C13). |

# 2 Model

Table 2: Results of the model, each with its statement and the Lean theorems that check it.

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M1 | Representation (P1). | $\Delta(m) = V\int \varphi\, dH_m = V\,\mathbb{E}[\varphi(M_m)]$, with $M_m \sim H_m$. | `EntryContestP1P2.two_max_sum`, `EntryContestP1P2.p1_representation`, `EntryContestP1P2.Delta_eq_expectation` | p. 15 |
| M2 | Dominance (P2). | For $s \geq 0$ and $\mu \leq 1$, $F \leq G$ and $\Delta(m) \geq 0$. For $\mu < 1$ and $\Pr(s > 0) > 0$, $F < G$ on some $[q, u)$ with $q < u$. | `EntryContestP1P2.p2_fosd`, `EntryContestP1P2.p2_strict_interval`, `EntryContestP1P2.Delta_nonneg_from_primitives` | p. 16 |
| M3 | Step identity (P3). | For $m+2 \leq Q$, $\Delta(m+1) - \Delta(m) = -\frac{V}{2}\int \varphi^2\, dK_m$. | `EntryContestStep.three_max_sum_real`, `EntryContestStep.step_identity`, `EntryContestModel.Delta_step` | p. 16 |
| M4 | Monotonicity. | $\Delta(m+1) \leq \Delta(m)$ for $V \geq 0$, with $<$ when $V > 0$ and $\int \varphi^2\, dK_m > 0$. | `EntryContestModel.Delta_step_nonpos`, `EntryContestModel.DeltaUpTo_step`, `EntryContestEq.Delta_step_strict` | p. 16 |
| M5 | Threshold entry (P5). | For $\kappa_0 \leq \cdots \leq \kappa_{Q-1}$ and $\Delta$ non-increasing, $\kappa_j \leq \Delta(j)$ and $i \leq j$ give $\kappa_i \leq \Delta(i)$. | `EntryContest.single_crossing`, `EntryContestModel.p5_threshold_from_primitives` | p. 17 |
| M6 | Count invariance. | Every equilibrium $S$, with $\kappa_i \leq \Delta(\lvert S\rvert - 1)$ for $i \in S$ and $\Delta(\lvert S\rvert) < \kappa_j$ for $j \notin S$, has $\lvert S\rvert = k^*$. | `EntryContestEq.count_invariance_fin`, `EntryContestMM.exists_mem_ge`, `EntryContestMM.exists_not_mem_le` | p. 17 |
| M7 | Identities pinned. | If $\Delta(k^* - 1) < \kappa_{k^*}$, the assortative set $\{0, \ldots, k^* - 1\}$ is the only equilibrium. | `EntryContest.members_below_kstar`, `EntryContestEq.assortative_unique` | p. 17 |
| M8 | Assortative selection (N4). | $\{0, \ldots, k^* - 1\}$ is an equilibrium; $\sum_{i<k} \kappa_i \leq \sum_{i \in S} \kappa_i$ for $\lvert S\rvert = k$; with identical score laws, payoff sums of equal-size equilibria differ by their cost sums. | `EntryContestEq.assortative_is_equilibrium`, `EntryContestEq.assortative_min_cost`, `EntryContestEq.prize_term_anonymous`, `EntryContestEq.payoff_gap_anonymous` | p. 17 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M9 | Benchmark (P6). | With the incumbent investing and $s \geq 0$ atomless, $\Delta(m) = V/(m+2)$ at $\mu = 0$ for every $Q$, and $\Delta(m) \to V/(m+2)$ as $\mu \to 0$. | `EntryContestP6P7.integral_iid`, `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p6_evaluation`, `EntryContestP6P7.p6_at_zero`, `EntryContestP6P7.tendsto_cdf_investorLaw`, `EntryContestP6P7.tendsto_cdf_nonInvestorLaw`, `EntryContestP6P7.p6_limit` | p. 18 |
| M10 | Saturation (P7). | $\Delta(m) \leq V/(m+1)$ for every $Q$, and an entrant at index $m$ has $(m+1)\kappa \leq V$. | `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p7_Delta_le`, `EntryContestP6P7.p7_cap`,`EntryContestP6P7.p7_count` | p. 18 |
| M11 | Prize and fee (P8). | $k^*$ is non-decreasing in $V$ and, for $u$ non-decreasing, non-increasing in $c$. | `EntryContestEq.kstar_mono`,`EntryContestEq.Delta_mono_prize`, `EntryContestMM.Delta_scale`,`EntryContestEq.kappa_mono_fee` | p. 18 |
| M12 | Margin condition. | For sorted profiles $w$ and $w'$, $w_{k^*-1} \leq w'_{k^*-1}$ gives $k^* \leq k'^*$, and $w'_{k^*} \leq w_{k^*}$ gives $k'^* \leq k^*$. | `EntryContestSpread.enters_down`, `EntryContestSpread.count_ge_of_enters`, `EntryContestSpread.count_le_of_fails`, `EntryContestSpread.margin_rise`, `EntryContestSpread.margin_fall` | p. 19 |
| M13 | Pivot spreads (P9-gen, P9). | For a pivot-spread $T$ about $x_0$, $k^*$ weakly rises when $w_{k^*-1} \geq x_0$ and weakly falls when $w_{k^*-1} \leq x_0$. The linear spread about the mean is one, and keeps the mean. A change that keeps the total and majorizes $w$ can lower $k^*$ with $w_{k^*-1}$ above the mean. | `EntryContestSpread.pivot_rise`, `EntryContestSpread.pivot_fall`, `EntryContestSpread.spread_isPivot`, `EntryContestMM.mm2a`, `EntryContestSpread.mps_lowers_count` | p. 19 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M14 | The band. | With $d = w' - w$, the signs of $d$ leave the direction of $k^*$ open only when $d_{k^*-1} < 0 < d_{k^*}$. A pivot-spread never does so. Inside the band two changes that keep the total and majorize $w$, with the same signs at every rank, move $k^*$ in opposite directions for every small $\mu > 0$. | `EntryContestSpread.branch_of_not_band`, `EntryContestSpread.not_band_of_single_crossing`, `EntryContestSpread.pivot_single_crossing`, `EntryContestSpread.mps_band_signs`, `EntryContestSpread.mps_witness_counts`, `EntryContestSpread.mps_band_both_directions`, `EntryContestSpread.mps_band_both_directions_positive` | p. 19 |
| M15 | Strictness under a spread (P9). | If some entrant lies below $x_0$ at $\lambda = 1$ and $\kappa(w) \to \infty$ as $w \downarrow c$, $k^*(\lambda)$ is non-increasing in $\lambda \geq 1$ and $k^*(\lambda) < k^*(1)$ above some $\bar{\lambda}$. | `EntryContestKappa.exit_below_pivot`, `EntryContestKappa.enters_downward`, `EntryContestKappa.p9_strict`, `EntryContestKappa.p9_monotone`, `EntryContestKappa.p9_strict_model` | p. 20 |
| M16 | Competitor number (P-MU). | For $Q \geq 1$ and every $C$, $D(Q) = -V\big(\int W_Q\,dF - \int W_Q\,dG\big)$, so for $V > 0$, $D(Q) > 0$ exactly when $\int W_Q\,dF < \int W_Q\,dG$. The kernel $g^n(1-g)$ rises on $[0, \frac{n}{n+1}]$ and falls on $[\frac{n}{n+1}, 1]$. | `EntryContestPMU.weight_eq`, `EntryContestPMU.pmu_identity`, `EntryContestPMU.pmu_sign_iff`, `EntryContestPMU.kernel_rises`, `EntryContestPMU.kernel_falls`, `EntryContestPMU.kernel_max` | p. 20 |
| M17 | Rise then fall in $Q$. | If $dF - dG \leq 0$ on $(-\infty, x_0]$ and $dF - dG \geq 0$ on $(x_0, \infty)$, and $V \geq 0$, then $D(Q_2) \leq G(x_0)^{Q_2 - Q_1} D(Q_1)$ for $1 \leq Q_1 \leq Q_2$, and $\min\big(\Delta_{Q_1}(0), \Delta_{Q_3}(0)\big) \leq \Delta_{Q_2}(0)$ for $1 \leq Q_1 \leq Q_2 \leq Q_3$. | `EntryContestPMU.weight_shift`, `EntryContestPMU.pmu_single_crossing`, `EntryContestPMU.pmu_step_nonpos_persists`, `EntryContestPMU.quasiconcave_of_steps`, `EntryContestPMU.pmu_quasiconcave` | p. 20 |
| M18 | Uniform base score. | For $r$ uniform on $[0,1]$, $s \geq 0$, $0 < \mu \leq 1$, $V \geq 0$ and every $C$, the hypothesis of M17 holds at $x_0 = \mu$, and $D(Q) \geq 0$ for $Q \geq 1$, so $\Delta_Q(0)$ is non-decreasing in $Q \geq 1$. | `EntryContestPMU.uniform_left`, `EntryContestPMU.uniform_right`, `EntryContestPMU.pmu_quasiconcave_uniform`, `EntryContestRefute.pmu_step_nonneg_uniform`, `EntryContestRefute.Delta_mono_uniform` | p. 21 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M19 | A class with rise then fall. | For $s \in \{0,1\}$ with $\Pr(s=0)=p$, $0<\mu<1$, $V \geq 0$, $\Pr(r \leq 0)=0$ and $\Pr(r \in B) \leq \Pr(r \in B-c)$ whenever $c>0$ and $B \subseteq (c,\infty)$, the hypothesis of M17 holds at $x_0 = 1-\mu$, and $\Delta_Q(0)$ is quasi-concave in $Q \geq 1$ for every $C$. A density of $r$ non-increasing on $(0,\infty)$ gives the shift condition, and the smaller of two uniform draws has one. | `EntryContestShift.left_of_lift`, `EntryContestShift.right_of_lift`, `EntryContestShift.quasiconcave_of_shiftMono`, `EntryContestShift.shiftMono_of_density`, `EntryContestShift.lawA_shiftMono`, `EntryContestShift.witness_quasiconcave` | p. 21 |
| M20 | The fall, exhibited. | For $r$ the smaller of two uniform draws, $\mu = \frac{3}{4}$, $s$ as in M19 and $C = F$, $D(1)$ and $D(2)$ are quadratics in $p$ (in the proof). For $V>0$, $p=\frac{1}{2}$ gives $\Delta_1(0) < \Delta_2(0)$ and $\Delta_3(0) < \Delta_2(0)$, and $p=\frac{1}{4}$ gives $\Delta_2(0) < \Delta_1(0)$. | `EntryContestFall.stepQ_one`, `EntryContestFall.stepQ_two`, `EntryContestFall.rise_then_fall`, `EntryContestFall.fall_from_one` | p. 21 |
| M21 | When the gain must fall. | With $s$ and $r$ as in M19, $t = 1-\mu$ and $G_t$ the law of $\mu r + t$, if $t < a < b$, $G(t) < G(a)$, $G(b) < 1$, $C(a) > 0$ and $G\big((a,b]\big) < G_t\big((a,b]\big)$, then for $V>0$ and $p<1$ some $Q_0$ has $D(Q)<0$ for every $Q \geq Q_0$. The base law of M20 meets this on $(\frac{1}{2}, \frac{5}{8}]$ for every $p<1$, and a uniform base score never does. | `EntryContestMustFall.must_fall_general`, `EntryContestMustFall.shift_dominates`, `EntryContestMustFall.stepQ_bern`, `EntryContestMustFall.must_fall`, `EntryContestMustFall.witness_must_fall`, `EntryContestMustFall.unif_shiftMono`, `EntryContestMustFall.uniform_never_gains` | p. 22 |
| M22 | Dominance does not sign $D(Q)$. | With $C = F$, the pairs $(F,G) = \big(x, 1-(1-x)^2\big)$ and $(x^2, x)$ on $[0,1]$ both have $F \leq G$ and no atoms. The first has $D(1) = \frac{V}{60}$ and $D(2) = \frac{V}{420}$, and the second has $D(Q) = -\frac{VQ}{(Q+2)(Q+3)(Q+4)}$. | `EntryContestPMUWitness.pairA_fosd`, `EntryContestPMUWitness.pairB_fosd`, `EntryContestPMUWitness.pairA_step_one`, `EntryContestPMUWitness.pairA_step_two`, `EntryContestPMUWitness.pairB_step`, `EntryContestPMUWitness.fosd_does_not_sign` | p. 22 |
| M23 | Burden-monotonicity. | $\kappa(w) = u(w) - u(w-c)$ is strictly decreasing on $w > c$ when $u$ is strictly concave on $(0,\infty)$, or when $u'(w) < u'(w-c)$ for every $w > c$. A convex stretch of $u$ longer than $c$ rules it out. Log and CRRA utility ($\gamma > 0$, $\gamma \neq 1$) satisfy it. | `EntryContestKappa.burden_strictAnti_of_strictConcave`, `EntryContestBM.burden_strictAnti_of_deriv`, `EntryContestBM.not_burden_of_convex_stretch`, `EntryContestKappa.log_burden_strictAnti`, `EntryContestKappa.crra_strictConcave`, `EntryContestKappa.crra_burden_strictAnti` | p. 22 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M24 | Weaker than concavity. | For every $c > 0$ some strictly increasing $u$, not concave on $(0,\infty)$, has $\kappa$ strictly decreasing on $(c,\infty)$ and $\kappa(w) \to \infty$ as $w \downarrow c$. The parameters of the construction form an open set, and its convex stretch can have any length below $c$. | `EntryContestBM.rampU_burden`, `EntryContestBM.rampU_kappa_diverges`, `EntryContestBM.rampU_strictConvex`, `EntryContestBM.rampU_not_concave`, `EntryContestBM.admissibleParams_open`, `EntryContestBM.rampU_admissible`, `EntryContestBM.admissible_width`, `EntryContestBM.bm_strictly_weaker` | p. 23 |
| M25 | Divergence at the fee. | For $u$ continuous at $c$, $\kappa(w) \to \infty$ as $w \downarrow c$ exactly when $u(x) \to -\infty$ as $x \downarrow 0$. Log utility and CRRA with $\gamma > 1$ satisfy it, and CRRA with $\gamma < 1$ does not. | `EntryContestKappa.kappa_diverges_iff`, `EntryContestKappa.not_diverges_of_continuousWithinAt_zero`, `EntryContestKappa.log_kappa_diverges`, `EntryContestKappa.crra_kappa_diverges`, `EntryContestKappa.crra_kappa_not_diverges` | p. 23 |
| M26 | Anonymity. | If every entrant's score has the law $F$, challenger $i$'s gain from entering against the entrants $T$ is $\Delta(\lvert T\rvert)$, whoever they are. | `EntryContestAnon.profile_mem`, `EntryContestAnon.profile_not_mem`, `EntryContestAnon.cdf_profile`, `EntryContestAnon.gain_anonymous` | p. 23 |
| M27 | The anonymity boundary. | If the law of $s_j$ varies with $j$, the gain depends on who the other entrants are. With ability falling in wealth, an equilibrium of one entrant and an equilibrium of two coexist for every small $\mu > 0$. | `EntryContestAnon.winProb_zero_entrant`, `EntryContestAnon.winProb_zero_outsider`, `EntryContestAnon.gain_zero`, `EntryContestAnon.kappa3_values`, `EntryContestAnon.gains3`, `EntryContestAnon.anon_witness_zero`, `EntryContestAnon.gain_tendsto`, `EntryContestAnon.anon_witness_positive`, `EntryContestAnon.anon_witness_exists` | p. 24 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M28 | Weight on talent. | With the incumbent investing, $s \geq 0$, $r$ atomless and $V \geq 0$, for $0 < \mu' \leq \mu$ and every $Q$ and $m$, $\Delta(m;\mu) \leq \Delta(m;\mu')$, and with fixed costs $k^*$ is non-increasing in $\mu$. | `EntryContestMu.Delta_normalise`, `EntryContestMu.cdf_payerLaw_antitone`, `EntryContestMu.first_integral`, `EntryContestMu.Delta_payer_mono`, `EntryContestMu.Delta_antitone_mu`, `EntryContestMu.kstar_antitone_mu` | p. 24 |
| M29 | The boundary, aligned. | With ability rising in wealth, an equilibrium of one entrant and an equilibrium of two coexist at $\mu = 0$ and for every small $\mu > 0$, with $s_j$ the largest of 12, 10 and 9 uniform draws, the incumbent's of one, and $\kappa$ from $u(x) = -1/x$, $c = 1$, $w = (\frac{43}{20}, \frac{52}{25}, \frac{83}{40})$. | `EntryContestAligned.kappaA_values`, `EntryContestAligned.ability_with_wealth`, `EntryContestAligned.costs_ordered`, `EntryContestAligned.gainsA`, `EntryContestAligned.aligned_witness_zero`, `EntryContestAligned.aligned_witness_positive`, `EntryContestAligned.aligned_witness_exists`, `EntryContestAligned.control_window` | p. 24 |
| M30 | The incumbent as a challenger. | Let the incumbent be player 0 of a game of $Q+1$ players with $\kappa_0 \leq \kappa_j$ and gain $g(m) = \Delta(m)$ at $C = G$; her being in gives a challenger $g(m+1)$. An equilibrium with 0 out and some challenger in exists iff $1 \leq k^* \leq Q$, $g(k^*) < \kappa_0$ and $\kappa_{k^*} \leq g(k^*-1)$; $\{0,\ldots,k^*-1\}$ is an equilibrium in every case, and if $g(k^*-1) < \kappa_{k^*}$ then 0 is in every non-empty equilibrium. | `EntryContestInc.Delta_incumbent_in_eq`, `EntryContestInc.symGain_step`, `EntryContestInc.incumbent_out_iff`, `EntryContestInc.incumbent_out_not_unique`, `EntryContestInc.incumbent_in_of_pinned`, `EntryContestInc.band_witness`, `EntryContestInc.control_no_band` | p. 25 |

| ID | Claim | Statement | Lean | Proof |
|---|---|---|---|---|
| M31 | The paying class's share. | With the incumbent investing and $k$ of $Q$ challengers paying, the prize goes to a payer with probability $\xi(k) = \int G^{Q-k}\, d(F^{k+1})$. For $F$ atomless, $\xi$ is non-decreasing in $k$ and $\xi(Q) = 1$; for $F = G$, $\xi(k) = \frac{k+1}{Q+1}$; for $s \geq 0$ and $r$ atomless, $\xi(k)$ is non-increasing in $\mu > 0$, and with fixed costs so is $\xi(k^*)$. | `EntryContestShare.payerShare_full`, `EntryContestShare.payerShare_eq_one_sub`, `EntryContestShare.payerShare_monotone`, `EntryContestShare.payerShare_at_equal`, `EntryContestShare.payerShare_normalise`, `EntryContestShare.payerShare_antitone_mu`, `EntryContestShare.payerShare_kstar_antitone_mu`, `EntryContestShare.control_share_needs_nonneg` | p. 25 |

# 3 Literature

Table 3: Papers the model or the conclusions depend on, each with a verbatim quote.

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L1 | Lazear and Rosen [1981] | Wealth sorting into a tournament is in print, by example and through declining absolute risk aversion, a condition on $u'''$. M23 sorts through $\kappa$ alone, which $u'' < 0$ gives. | "Individuals will self-select the payment scheme in accordance with their wealth." | 855 | M5, M23 |
| L2 | Levin and Smith [1994] | With identical potential entrants, pure strategies leave who enters unexplained, and mixed entry makes the count random. | "The process by which potential bidders divide into these two groups is not explained" | 586 | M6, M7 |
| L3 | Morgan et al. [2012] | With identical players the entrants' identities are not pinned, and sequential entry orders them. | "the identity of the players choosing to opt into the contest is not uniquely determined" | 441 | M6, M7 |
| L4 | Moreno and Wooders [2011] | Under private entry costs every buyer compares her cost with one threshold, and the count is binomial. Here the threshold depends on the index. | "indicating the maximum entry cost for which the buyer enters the auction" | 319–320 | M5, M6 |
| L5 | Fullerton and McAfee [1999] | With heterogeneous types the efficient entry equilibrium admits the lowest-cost firms. Profits depend on rivals' types, and entry equilibria of sizes 2 and 3 coexist (`FullertonMcAfee.two_sizes`), so count invariance needs anonymity. | "entry of the m lowest-cost firms is an equilibrium" | 580 | M6, M7, M26 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L6 | Fu et al. [2015] | The rent-accounting bound behind M10, for identical agents with mixed entry, capping an expected count. | "the rent to be dissipated in the contest cannot exceed" | 397 | M10 |
| L7 | Fu and Lu [2010] | Total effort equals the budget less the entrants' total entry cost, in a contest chosen by a designer. *Unverified:* the working paper (MPRA 945) was read, and the *Economic Inquiry* version is not checked. | "the equilibrium total effort is given by the difference between the total budget of the contest organizer and the total entry costs incurred by participating contestants, regardless of the contest technology." | 10 | M10 |
| L8 | Ryvkin and Drugov [2020] | The order-statistic kernel, and that no prediction in the number of players holds for every noise law. M16 to M22 are the entry-gain instance. | "There is not a single prediction, either for individual or aggregate effort, that cannot be reversed for at least some distribution of noise." | 1589 | M16, M17, M22 |
| L9 | Costrell and Loury [2004] | A spread's effect signed by the position of one marginal quantile, which is fixed by technology there and endogenous in M12. *Unverified:* a draft was read, and the published version is not checked. | "completely characterize the technology" | 6 | M12, M13 |
| L10 | Hopkins and Kornienko [2010] | The dispersive order as the order for inequality comparisons, quoting Shaked and Shanthikumar (2007). | "A variable with distribution $F$ is said to be smaller in the dispersive order (or less dispersed) than a variable with a distribution $G$" | 128 | M13, M14 |
| L11 | Lewis and Thompson [1981] | The same order, defined earlier by quantile spacing, which with a crossing point gives the pivot-spreads of M13. | "We call the relationship defined by (1.1), (1.2) an ordering in dispersion of $G$ and $H$" | 77 | M13, M14 |
| L12 | Schroyen and Treich [2016] | The privilege-contest payoff, with a prize valued apart from wealth, and an intensive-margin effect of a spread that turns on $u'''$. *Unverified:* the working paper (No. 16-699) was read, and the *Games and Economic Behavior* version is not checked. | "and therefore its marginal value is independent of the level of wealth" | 8 | M13, M23 |
| L13 | Cole et al. [1992] | A more compact income distribution raises savings through matching on wealth. The direction matches the below-pivot branch of M13, and the mechanism differs. | "the agent from the economy with the more compact income distribution will tend to have a higher savings rate" | 1103 | M13 |
| L14 | Weber [1978] | Property excludes the poor from competing, and does so through marginal utility. Here that channel is the fee's utility cost $\kappa$, which falls with wealth (M23), so entry is richest-first (M5). | "The mode of distribution, in accord with the law of marginal utility, excludes the non-wealthy from competing for highly valued goods" | 927 | M5, M23 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L15 | Weber [1978] | Class and status are distinct, and money does not by itself confer status. Here wealth sets only the cost of entry, and the prize goes by rank on draws that do not depend on wealth. Weber gives this anonymity to the market and not to the status order (p. 936), so the contest is market-like. | "Money and an entrepreneurial position are not in themselves status qualifications, although they may lead to them" | 306 | M26, M27 |
| L16 | Weber [1978] | Closure, participation subject to a condition. The fee is such a condition, and M4 gives entrants an interest in it, since each further entrant lowers the gain. No agent sets the fee here. | "participation of certain persons is excluded, limited, or subjected to conditions" | 43 | M4, M10 |
| L17 | Laurison and Friedman [2016] | Access to the traditional professions is more closed than to management, which earns more, so closure does not follow the prize. Their main finding, a pay gap after entry, is outside the model, which stops at entry. *Unverified:* the accepted version was read, and the published pagination is not checked. | "the higher professions remain significantly more elitist in terms of restricting access for those from working class backgrounds" | 13 | M5, M11, M28 |
| L18 | Wegener [1992] | A survey of prestige in stratification theory, which separates closure from a hierarchy of positions and finds no theory that joins them. The question here is narrower, namely how wealth differences decide who competes for positions open to every entrant on one score. | "None of these theories is able to solve the problem of how theoretically to merge the idea of social closure with that of a hierarchy of positions." | 253 | M5, M26 |
| L19 | Tilly [1998] | In opportunity hoarding a bounded set keeps access to a valuable resource. The outcome is the same here, the richest $k^*$ taking the prize with probability $\xi(k^*)$. The means differ, since in Tilly the network acts and here no one does. *Unverified:* one page read, and the edition is not confirmed. | "If members of a network acquire access to a resource that is valuable, renewable, subject to monopoly, supportive of network activities, and enhanced by the network's modus operandi, network members regularly hoard access to the resource, creating beliefs and practices that sustain their control." | 154 | M8, M31 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L20 | Morgan et al. [2022] | With an effort choice, contestants drop out as ranking of the costly choice becomes more accurate, once effort competes the rents away. Their accuracy is a fall in $\mu$ here, so the sign is the opposite of M28; entry by a fixed fee has no effort margin. *Unverified:* the 2018 working paper was read, and the *Journal of Economic Theory* version is not checked. | "at a critical point, competition becomes so intense that contestants start 'dropping out,' i.e., they put in zero effort" | 4 | M28 |
| L21 | Shaked [1982] | The dispersive order holds exactly when every shift of one distribution function crosses the other at most once, from below (Theorem 2.1). This is the crossing in the pivot-spreads of M13. He credits the relation to Saunders and Moran (1978) and its name to Lewis and Thompson (L11). | "In the first pair of theorems it is shown that order in dispersion is equivalent to the condition that some functions, determined by the pair of the underlying distributions, change sign at most once." | 311 | M13, M14 |
| L22 | Drugov and Ryvkin [2020] | Effort falls as noise becomes more dispersed, for any prize schedule (Proposition 1), the sign of M28. With free entry, the less noisy tournament has weakly fewer entrants (Section 5.3), the opposite sign, because effort rises as noise falls. *Unverified:* the working paper (NES WP 256) was read, and the *Journal of Economic Theory* version is not checked. | "the dispersive order is a necessary and sufficient condition for the ranking of equilibrium efforts in general" | 10 | M28 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L23 | Mathews and Namoro [2008] | Two events pay different prizes, and a strong and a weak contestant each join one. The strong contestant can join the event with the smaller prize and the weak one the event with the larger prize (their allocation c), so a smaller prize can attract the stronger field. In M11 a larger $V$ never lowers the number who pay the fee, because $V$ raises the gain from entering at every count. Their contestants choose between events at no entry cost, the number of strategic contestants is fixed at two, and wealth does not appear. *Unverified:* the working paper of 29 March 2005 was read, and the published version (*Journal of Economics* 2008) is not checked. | "Event 2 is able to attract a field of higher quality by offering smaller prizes." | 13 | M8, M11 |
| L24 | Corcoran [1984] | Free entry into a Tullock lottery among identical players: players enter while the return on the lottery exceeds the return on other projects and exit while it falls short. At the long-run count the total spent equals the discounted prize, and for $r > 1$ the count $r/(r-1)$ does not depend on the prize. Expenditure in his model is proportional to the prize. Here the fee $\kappa$ does not depend on $V$, $k^*$ is non-decreasing in $V$ (M11), and an entrant at index $m$ has $(m+1)\kappa \leq V$ (M10). His players are identical, their number is a real number, and wealth does not appear. | "If the rate of return obtainable on other projects is less than that offered by the rent-seeking option then players will enter; if greater, players will exit." | 91 | M5, M10, M11 |

| ID | Paper | What we rely on | Quote | Page | Model rows |
|---|---|---|---|---|---|
| L25 | Fryer and Loury [2010] | Agents choose a binary effort whose cost is drawn from their group's distribution, productivity is a noisy function of effort, and slots in a hierarchy go top-down through a market, so those who do not invest still compete. Under laissez-faire the group with higher costs sits uniformly lower and, with two jobs, holds fewer slots. This is the class gap of M31 for groups, in a continuum with priced slots; there is no count of payers, no fixed prize and no weight on talent. *Unverified:* the working paper (NBER WP 16568) was read, and the *Journal of Political Economy* version (2013) is not checked. | "So, a disadvantaged group would be underrepresented (holding a fraction of the slots that is less than their fraction in the population), since their effort costs are higher (Assumption 1), and effort is productive (Assumption 2)." | 8 | M31, C10, C13 |
| L26 | Fu and Lu [2009] | Identical contestants choose a continuous investment that lowers their cost of effort, and then all compete in a Tullock contest. The paper finds investment and effort non-monotonic in the number of contestants. Within the symmetric pure-strategy equilibria it studies, that does not hold: at its parameter zero investment is not an equilibrium for $K \geq 4$, and its candidate for $K = 3$ never is one (record in `lit/fu_lu_2009`). There is no binary investment and no one who does not invest, so nothing corresponds to the gain of M17 and M18. *Unverified:* the authors' manuscript of February 2009 was read, and the *Economics Letters* pages are not checked. | "Positive equilibrium investment emerges when the number of contestants is small, while zero investment occurs in equilibrium when the number of contestants is large." | 2 | M17, M18, C7 |

A paper cited in our other documents and not read is not in the table, and every claim made about it is unverified. It is Corcoran and Karels [1985].

# 4 Conclusions

The conclusions come in two tables. Table 4 holds the claims that rest on the model rows and on papers read in full. Table 5 holds the claims that something is new. Each of those waits on a paper not yet read in full, or on a comparison not yet done, and says which. The novelty reading is pass 7 of `TODO.md`.

Table 4: Conclusions that rest on the model rows and on papers read in full.

| ID | Statement | Rests on |
|---|---|---|
| C1 | For a fixed prize awarded by rank, a spread of wealth changes the number of entrants in a direction set by one comparison, the marginal entrant against the pivot of the spread, and the rule needs only that the fee's utility cost falls with wealth. In the privilege contest of Schroyen and Treich the effect of a spread on effort turns on $u'''$, so the entry margin is what frees the sign from the third derivative. Their paper was read in its working-paper version (L12). | M12, M13, M23, M24, L12 |
| C2 | The rule reads the wealth profile at two ranks only, the marginal entrant and the first outsider, and it holds for any change of profile, mean-preserving or not. Where the marginal entrant becomes poorer and the first outsider richer, the signs of the change settle nothing, even for mean-preserving spreads, and the mean cannot stand in for the pivot. Costrell and Loury sign a spread by one marginal quantile fixed by technology, and here the margin is the count itself, which moves with the profile. Their paper was read as a draft (L9). | M12, M13, M14, L9 |
| C3 | Every pure-strategy equilibrium has the same number of entrants because the gain depends on that number alone. When the gain depends on who enters, the count is not pinned. This happens in the model when ability falls with wealth, and in the entry stage of Fullerton and McAfee, where profits depend on rivals' costs. | M6, M26, M27, L5 |
| C4 | Under complete information, heterogeneous entry costs pin the count without the devices that models with identical entrants use (mixed entry, sequential arrival, private random delays), and the $k^*$ richest entering is always an equilibrium. Who enters is pinned only under the further condition of M7. | M5, M6, M7, M8, L2, L3, L4 |
| C9 | With the fee and the prize fixed, the number of entrants does not rise as the rule weighs talent more, for every law of talent and of the bought component, when the incumbent is scored by the same rule. For an incumbent protected from the rule, whose law does not move with the weight, the argument does not apply (Appendix D). The sign belongs to entry by a fixed fee in which non-payers still compete. With an effort choice it is the opposite (L20, L22), because effort rises with the accuracy of the ranking until entry earns nothing, while a fixed fee does not rise (Appendix D, "No effort margin"). | M28, L20, L22 |
| C10 | The share of the prize taken by those who pay is set by the count, so it moves with the prize, the fee and the wealth profile as the count does, and it does not rise as the rule weighs talent more. That a group with higher investment costs holds fewer positions is Fryer and Loury's result under laissez-faire, for identity groups in a continuum (L25). The outcome is the opportunity hoarding of Tilly [1998], a bounded group keeping access to a valuable resource, but here no one hoards, and the fee and the wealth profile produce it. Laurison and Friedman [2016] find access to the traditional professions more closed than to management, which pays more. Read as shares, a larger prize alone does not lower the share, so the prize does not account for that pattern, and a rule weighing the bought component more is one difference that would. | M11, M28, M31, L17, L19, L25 |
| C11 | An incumbent scored by the same rule, with the lowest cost, sits out while a challenger enters only as one of several equilibria with the same count, inside the band of M6, and never when identities are pinned. | M6, M7, M30 |
| C14 | The first entrant's gain is $V\,\mathbb{E}[\varphi(M)]$, with $M$ the best rival's score (M1). This is the form of the marginal benefit of Ryvkin and Drugov [2020], the noise density at the best rival's shock, with the score gap $\varphi$ in place of the density. Their Karlin step, with the law of $M$ in place of $F^{k-1}$, gives M17, no interior minimum in the number of challengers when $\varphi$ is single-peaked. Their remark that with many players the comparative statics follow the upper tail is what M21 makes precise for the gain. | M1, M17, M21, L8 |

Table 5: Claims pending verification. Each says what it waits on.

| ID | Statement | Rests on |
|---|---|---|
| C5 | The sign rule of C1 is new. None of the entry papers read signs the effect of a spread of wealth, or of entry costs, on the number of entrants at a fixed number of potential entrants. *Pending:* Corcoran and Karels [1985], an entry paper that is not read. | M13, L2, L3, L4, L5, L6, L7, L23, L24 |
| C6 | Count invariance across pure-strategy equilibria with heterogeneous entry costs is new. The entry papers read have identical entrants (Levin and Smith, Morgan, Orzen and Sefton, Fu, Jiao and Lu, Fu and Lu, Corcoran), a random count under private costs (Moreno and Wooders), or, in Fullerton and McAfee, a count that differs across equilibria. Mathews and Namoro fix the number of strategic entrants at two. *Pending:* Corcoran and Karels [1985], not read. | M6, L2, L3, L4, L5, L6, L7, L23, L24 |

| ID | Statement | Rests on |
|---|---|---|
| C7 | Beyond what C14 credits to Ryvkin and Drugov, two statements about the gain from entry are new. With a uniform base score the gain never falls in the number of challengers, because the peak of the score gap sits at the top of the outsiders' support, where the kernel vanishes, which it cannot do in their tournament (M18). Primitive conditions on the base score and the bought component make the score gap single-peaked (M19). *Pending:* no paper known. The one candidate that the search of 13 October 2026 found, Fu and Lu [2009], is read and is not a precedent (L26); the claim rests on that search. | M18, M19, L8, L26 |
| C8 | That two ranks suffice to sign the change in the count is a sharpening that the assignment literature does not have. *Pending:* the published version of Costrell and Loury, read here as a draft. | M12, L9 |
| C12 | The sign of M28 on the entry margin is new. The entry margins read have the opposite sign, more noise bringing more entrants, because effort rises as noise falls and competes the rents of entry away (L20; Drugov and Ryvkin's free-entry extension, L22). M28's sign is that of their intensive margin, effort falling as noise becomes more dispersed (L22), because paying for the bought component is the investment itself. The contrast with the number of challengers, which has no universal sign, is theirs across their two papers (L8, L22). Fryer and Loury conjecture, without deriving it, that a blind rule rationing slots at random lowers investment in both groups, the direction of M28 (L25). *Pending:* Coate and Loury [1993], where the informativeness of a test shapes a binary investment, and Fryer and Loury [2005], where a blind rule creates a pool of contestants who exert no effort; both found in the search of 13 October 2026 and not read. | M16, M28, L8, L20, L22, L25 |
| C13 | The share of the prize taken by those who enter, as a function of how many enter, and its comparative statics are new in the entry literature. In every entry model read, those who do not enter cannot win, so the share is not defined there; here outsiders compete on the base score. In Mathews and Namoro, non-strategic contestants fill the places that the strategic ones leave and can win, but they make no choice, the number of strategic contestants is fixed at two, and no share is analysed (L23). Outside the entry literature, Fryer and Loury have a binary investment in which those who do not invest still compete, and the group with higher costs holds fewer slots (L25). Their agents form a continuum and slots are priced, so they do not derive the payers' share as a function of how many pay, nor its comparative statics. *Pending:* Coate and Loury [1993] and Corcoran and Karels [1985], not read, and the published version of Fryer and Loury. | M31, L2, L3, L4, L5, L6, L7, L23, L24, L25 |

# A Proofs

**Notation.** Challenger $i$ scores $\sigma_i = \mu r_i + (1-\mu)s_i e_i$, with $r_i$ and $s_i \geq 0$ drawn independently of wealth. $F$ and $G$ are the score CDFs of an entrant and of an outsider, and $C$ is the incumbent's, each without atoms. With $m$ other entrants among the $Q$ challengers,

$$H_m = C F^m G^{Q-1-m}, \qquad K_m = C F^m G^{Q-2-m}, \qquad \varphi = G - F,$$

$$\Delta(m) = V\Big( \int H_m \, dF - \int H_m \, dG \Big), \qquad \kappa(w) = u(w) - u(w-c).$$

$H_m$ is the CDF of the best score facing a challenger, $M_m$ a random variable with that CDF, and $K_m$ the same CDF with one more challenger set aside, so that $H_{m+1} = K_m F$ and $H_m = K_m G$. Challengers are indexed $0, \ldots, Q-1$ in increasing order of cost, $\kappa_0 \leq \cdots \leq \kappa_{Q-1}$, which is decreasing order of wealth. Index $j$ enters when $\kappa_j \leq \Delta(j)$, and $k^*$ satisfies $\kappa_j \leq \Delta(j)$ for $j < k^*$ and $\Delta(k^*) < \kappa_{k^*}$ when $k^* < Q$. In the results on wealth profiles, rank $j$ has wealth $w_j$ with $w_0 \geq w_1 \geq \cdots$, and enters when $w_j > c$ and $\kappa(w_j) \leq \Delta(j)$, so a challenger whose wealth is at or below the fee does not enter. A prime marks the profile after a change and its count $k'^*$. When the number of challengers varies, $\Delta_Q(0)$ is the first entrant's gain among $Q$ challengers, with

$$D(Q) = \Delta_{Q+1}(0) - \Delta_Q(0), \qquad W_Q = C G^{Q-1}(1-G).$$

### Proof of M1

*Lean:* `EntryContestP1P2.two_max_sum`, `EntryContestP1P2.p1_representation`, `EntryContestP1P2.Delta_eq_expectation`.

For independent $X \sim A$ and $Y \sim B$ with $B$ atomless, $\Pr(X = Y) = 0$, so

$$\int B \, dA + \int A \, dB = \Pr(Y \leq X) + \Pr(X \leq Y) = 1.$$

With $(A,B) = (F,H_m)$ and $(A,B) = (G,H_m)$,

$$\int H_m\,dF = 1 - \int F\,dH_m, \qquad \int H_m\,dG = 1 - \int G\,dH_m,$$

so

$$\frac{\Delta(m)}{V} = \int (G-F)\,dH_m = \int \varphi\,dH_m = \mathbb{E}[\varphi(M_m)].$$

□

## Proof of M2

*Lean:* `EntryContestP1P2.p2_fosd`, `EntryContestP1P2.p2_strict_interval`, `EntryContestP1P2.Delta_nonneg_from_primitives`.

Let $X = \mu r + (1-\mu)s$ and $Y = \mu r$. For $\mu \le 1$ and $s \ge 0$, $X - Y = (1-\mu)s \ge 0$, so $\{X \le x\} \subseteq \{Y \le x\} \cup \{s < 0\}$ and

$$F(x) \le G(x) + \Pr(s < 0) = G(x), \qquad \Delta(m) = V \int \varphi\,dH_m \ge 0.$$

For $\mu < 1$,

$$\{s > 0\} \subseteq \{Y < X\} = \bigcup_{q \in \mathbb{Q}} \{Y \le q < X\},$$

so $\Pr(s > 0) > 0$ gives $q \in \mathbb{Q}$ with $\Pr(Y \le q < X) > 0$, and

$$G(q) \ge \Pr(X \le q) + \Pr(Y \le q < X) > F(q).$$

$G - F$ is right-continuous, so $G - F > 0$ on some $[q,u)$.

□

## Proof of M3

*Lean:* `EntryContestStep.three_max_sum_real`, `EntryContestStep.step_identity`, `EntryContestModel.Delta_step`.

Write $K = K_m$, so $H_{m+1} = KF$ and $H_m = KG$. For independent atomless draws $Z \sim K$, $X \sim A$, $Y \sim B$, exactly one is the largest,

$$\int AB\,dK + \int KB\,dA + \int KA\,dB = 1.$$

With $(A,B) = (F,F)$, $(G,G)$ and $(F,G)$,

$$\int KF\,dF = \tfrac{1}{2}\Big(1 - \int F^2 dK\Big), \quad \int KG\,dG = \tfrac{1}{2}\Big(1 - \int G^2 dK\Big), \quad \int KG\,dF + \int KF\,dG = 1 - \int FG\,dK,$$

so

$$\frac{\Delta(m+1) - \Delta(m)}{V} = \int KF\,dF - \int KF\,dG - \int KG\,dF + \int KG\,dG = -\tfrac{1}{2}\int (F-G)^2\,dK.$$

□

## Proof of M4

*Lean:* `EntryContestModel.Delta_step_nonpos`, `EntryContestModel.DeltaUpTo_step`, `EntryContestEq.Delta_step_strict`.

By M3 and $\varphi^2 \ge 0$,

$$\Delta(m+1) - \Delta(m) = -\tfrac{V}{2}\int \varphi^2\,dK_m \le 0 \quad (V \ge 0), \qquad < 0 \quad \Big(V > 0,\ \int \varphi^2\,dK_m > 0\Big).$$

Beyond $m = Q - 1$ the gain is set to $\Delta(Q-1)$, so $\Delta(m+1) \le \Delta(m)$ for every $m$.

□

## Proof of M5

*Lean:* `EntryContest.single_crossing`, `EntryContestModel.p5_threshold_from_primitives`.

For $i \le j$, by M4,

$$\kappa_i \le \kappa_j \le \Delta(j) \le \Delta(i).$$

□

## Proof of M6

*Lean:* `EntryContestEq.count_invariance_fin`, `EntryContestMM.exists_mem_ge`, `EntryContestMM.exists_not_mem_le`.

Let $S$ be an equilibrium and $k = |S|$. A $k$-set of indices has an element $i \ge k-1$, since $S \subseteq \{0,\dots,k-2\}$ gives $k \le k-1$, and omits some $j \le k$, since $\{0,\dots,k\} \subseteq S$ gives $k+1 \le k$. If $k > k^*$, then $k^* < Q$ and

$$\Delta(k^*) < \kappa_{k^*} \le \kappa_{k-1} \le \kappa_i \le \Delta(k-1) \le \Delta(k^*).$$

If $k < k^*$, then $j < Q$ and

$$\Delta(k) < \kappa_j \le \kappa_k \le \Delta(k).$$

Both are contradictions, so $k = k^*$.

□

## Proof of M7

*Lean:* `EntryContest.members_below_kstar`, `EntryContestEq.assortative_unique`.

A member $i$ of an equilibrium has $\kappa_i \le \Delta(k^*-1)$ by M6. If $i \ge k^*$,

$$\kappa_i \ge \kappa_{k^*} > \Delta(k^*-1),$$

so $i < k^*$. Then $S \subseteq \{0,\dots,k^*-1\}$ and $|S| = k^*$ give $S = \{0,\dots,k^*-1\}$.

□

## Proof of M8

*Lean:* `EntryContestEq.assortative_is_equilibrium`, `EntryContestEq.assortative_min_cost`, `EntryContestEq.prize_term_anonymous`, `EntryContestEq.payoff_gap_anonymous`.

(i) For $i < k^*$ and for $k^* \le j < Q$,

$$\kappa_i \le \kappa_{k^*-1} \le \Delta(k^*-1), \qquad \Delta(k^*) < \kappa_{k^*} \le \kappa_j.$$

(ii) Induct on $S$, adding its largest element $a$ last. With $T = S \setminus \{a\}$ and $T \subseteq \{0,\dots,a-1\}$, $|T| \le a$, so

$$\sum_{i<|T|+1} \kappa_i = \sum_{i<|T|} \kappa_i + \kappa_{|T|} \le \sum_{i\in T} \kappa_i + \kappa_a = \sum_{i\in S} \kappa_i.$$

(iii) With identical laws, the win probability of $i$ is $P^{\text{in}}(|S|) = \int CF^{|S|-1}G^{Q-|S|}\,dF$ for $i \in S$ and $P^{\text{out}}(|S|) = \int CF^{|S|}G^{Q-1-|S|}\,dG$ for $i \notin S$, so with $EU_i(S) = u(w_i) - \kappa_i \mathbf{1}\{i \in S\} + VP_i(S)$,

$$\sum_i EU_i(S) = \sum_i u(w_i) - \sum_{i\in S} \kappa_i + V\big(|S|P^{\text{in}}(|S|) + (Q-|S|)P^{\text{out}}(|S|)\big),$$

and the last term is the same for any two entrant sets of one size.

□

### Proof of M9

*Lean:* `EntryContestP6P7.integral_iid`, `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p6_evaluation`, `EntryContestP6P7.p6_at_zero`, `EntryContestP6P7.tendsto_cdf_investorLaw`, `EntryContestP6P7.tendsto_cdf_nonInvestorLaw`, `EntryContestP6P7.p6_limit`.

For $X_0, \dots, X_k$ independent with atomless CDF $F$,

$$\int F^k \, dF = \Pr\Big(X_0 = \max_{i \le k} X_i\Big) = \frac{1}{k+1}.$$

At $\mu = 0$ the entrant scores $s$ and the outsider 0, so $G = \mathbf{1}_{[0,\infty)}$ and $F(0) = \Pr(s \le 0) = 0$. With $C = F$,

$$\int H_m \, dF = \int F^{m+1} \, dF = \frac{1}{m+2}, \qquad \int H_m \, dG = H_m(0) = 0.$$

For $\mu \to 0$ let $X_\mu = \mu r + (1-\mu)s$ and $Y_\mu = \mu r$. For $s > 0$,

$$F_\mu(X_\mu) \to F_0(s), \qquad G_\mu(X_\mu) \to 1, \qquad F_\mu(Y_\mu) \to F_0(0) = 0,$$

with every integrand in $[0,1]$, so by dominated convergence

$$\int H_m \, dF_\mu \to \int F_0^{m+1} \, dF_0 = \frac{1}{m+2}, \qquad 0 \le \int H_m \, dG_\mu \le \mathbb{E}\big[F_\mu(Y_\mu)\big] \to 0.$$

□

### Proof of M10

*Lean:* `EntryContestP6P7.integral_cdf_pow`, `EntryContestP6P7.p7_Delta_le`, `EntryContestP6P7.p7_cap`, `EntryContestP6P7.p7_count`.

From $C \le 1$, $G \le 1$ and $\int H_m \, dG \ge 0$,

$$\Delta(m) \le V \int H_m \, dF \le V \int F^m \, dF = \frac{V}{m+1}.$$

An entrant at index $m$ with cost $\kappa$ has $\kappa \le \Delta(m)$, so

$$(m+1)\,\kappa \le (m+1)\,\Delta(m) \le V, \qquad m+1 \le \frac{V}{\kappa} \quad (\kappa > 0).$$

□

### Proof of M11

*Lean:* `EntryContestEq.kstar_mono`, `EntryContestEq.Delta_mono_prize`, `EntryContestMM.Delta_scale`, `EntryContestEq.kappa_mono_fee`.

Let $k^*$ and $k'^*$ be the counts for $(\kappa, \Delta)$ and $(\kappa', \Delta')$, with $\kappa'_j \le \kappa_j$ and $\Delta(j) \le \Delta'(j)$ for every $j$. If $k'^* < k^*$, then $k'^* < Q$ and

$$\Delta'(k'^*) < \kappa'_{k'^*} \le \kappa_{k'^*} \le \Delta(k'^*) \le \Delta'(k'^*),$$

so $k^* \le k'^*$. Write $\Delta_V$ for the gain at prize $V$. By M1 and M2, $\Delta_V(j) = V\,\Delta_1(j)$ with $\Delta_1(j) \ge 0$, and for $u$ non-decreasing,

$$V \le V' \;\Rightarrow\; \Delta_V \le \Delta_{V'}, \qquad c \le c' \;\Rightarrow\; u(w) - u(w-c) \le u(w) - u(w-c').$$

□

## Proof of M12

*Lean:* `EntryContestSpread.enters_down`, `EntryContestSpread.count_ge_of_enters`, `EntryContestSpread.count_le_of_fails`, `EntryContestSpread.margin_rise`, `EntryContestSpread.margin_fall`.

(i) The marginal entrant has $c < w_{k^*-1} \le w'_{k^*-1}$, so

$$\kappa(w'_{k^*-1}) \le \kappa(w_{k^*-1}) \le \Delta(k^*-1),$$

and for $i \le k^*-1$, with $w'_i \ge w'_{k^*-1} > c$,

$$\kappa(w'_i) \le \kappa(w'_{k^*-1}) \le \Delta(k^*-1) \le \Delta(i).$$

Every $i < k^*$ enters after the change, so $k^* \le k'^*$. (ii) For $k^* < Q$, rank $k^*$ fails before the change. If $w'_{k^*} > c$, then $c < w'_{k^*} \le w_{k^*}$ and

$$\kappa(w'_{k^*}) \ge \kappa(w_{k^*}) > \Delta(k^*),$$

so rank $k^*$ fails after the change and $k'^* \le k^*$.

□

## Proof of M13

*Lean:* `EntryContestSpread.pivot_rise`, `EntryContestSpread.pivot_fall`, `EntryContestSpread.spread_isPivot`, `EntryContestMM.mm2a`, `EntryContestSpread.mps_lowers_count`.

$T$ is non-decreasing, so $w'_j = T(w_j)$ is sorted, and

$$x_0 \le w_{k^*-1} \;\Rightarrow\; w_{k^*-1} \le T(w_{k^*-1}), \qquad w_{k^*} \le w_{k^*-1} \le x_0 \;\Rightarrow\; T(w_{k^*}) \le w_{k^*},$$

so M12 applies. For $T(w) = x_0 + \lambda(w - x_0)$ with $\lambda \ge 1$,

$$T(w) - w = (\lambda - 1)(w - x_0), \qquad \frac{1}{Q}\sum_j T(w_j) = \bar{w} + \lambda\left(\frac{1}{Q}\sum_j w_j - \bar{w}\right) = \bar{w} \quad (x_0 = \bar{w}).$$

With $c = 1$, $\kappa(w) = \frac{1}{w-1} - \frac{1}{w}$ and $\Delta(m) = \frac{1}{m+2}$,

$$w = \left(\tfrac{29}{10}, \tfrac{14}{5}, 2\right), \quad w' = \left(\tfrac{18}{5}, \tfrac{11}{5}, \tfrac{19}{10}\right), \quad \textstyle\sum_j w_j = \sum_j w'_j, \quad w'_2 \le w_2, \quad w'_2 + w'_1 \le w_2 + w_1,$$

$$\kappa(w) = \left(\tfrac{100}{551}, \tfrac{25}{126}, \tfrac{1}{2}\right), \quad \kappa(w') = \left(\tfrac{25}{234}, \tfrac{25}{66}, \tfrac{100}{171}\right), \quad \tfrac{1}{3}\textstyle\sum_j w_j = \tfrac{77}{30} < w_1,$$

so the counts are 2 and 1.

□

## Proof of M14

*Lean:* `EntryContestSpread.branch_of_not_band`, `EntryContestSpread.not_band_of_single_crossing`, `EntryContestSpread.pivot_single_crossing`, `EntryContestSpread.mps_band_signs`, `EntryContestSpread.mps_witness_counts`, `EntryContestSpread.mps_band_both_directions`, `EntryContestSpread.mps_band_both_directions_positive`.

Outside the band $d_{k^*-1} \ge 0$ or $d_{k^*} \le 0$, and M12 signs the change. If

$$i \le j,\ d_j > 0 \;\Rightarrow\; d_i \ge 0,$$

then $d_{k^*} > 0$ gives $d_{k^*-1} \ge 0$. A pivot-spread has this property, since $d_j > 0$ gives $w_j > x_0$ and then $w_i \ge w_j > x_0$. With $\kappa$ and $\Delta$ as in M13, the profiles

$$w = \left(4, \tfrac{14}{5}, \tfrac{11}{5}, 2\right), \quad w' = \left(\tfrac{41}{10}, \tfrac{27}{10}, \tfrac{13}{5}, \tfrac{8}{5}\right), \quad w'' = \left(\tfrac{23}{5}, \tfrac{9}{4}, \tfrac{9}{4}, \tfrac{19}{10}\right)$$

all sum to 11, with partial sums from the poorest up

$$\big(2,\tfrac{21}{5},7\big),\quad \big(\tfrac{8}{5},\tfrac{21}{5},\tfrac{69}{10}\big),\quad \big(\tfrac{19}{10},\tfrac{83}{20},\tfrac{32}{5}\big),$$

and

$$\kappa(w)=\big(\tfrac{1}{12},\tfrac{25}{126},\tfrac{25}{66},\tfrac{1}{2}\big),\quad \kappa(w')=\big(\tfrac{100}{1271},\tfrac{100}{459},\tfrac{25}{104},\tfrac{25}{24}\big),\quad \kappa(w'')=\big(\tfrac{25}{414},\tfrac{16}{45},\tfrac{16}{45},\tfrac{100}{171}\big)$$

against $\Delta=\big(\tfrac{1}{2},\tfrac{1}{3},\tfrac{1}{4},\tfrac{1}{5}\big)$. The counts are 2, 3 and 1, and $d'$ and $d''$ both have signs $(+,-,+,-)$. Every inequality is strict, so by M9 the counts hold for every small $\mu>0$.

□

## Proof of M15

*Lean:*
`EntryContestKappa.exit_below_pivot`, `EntryContestKappa.enters_downward`, `EntryContestKappa.p9_strict`, `EntryContestKappa.p9_monotone`, `EntryContestKappa.p9_strict_model`.

Let $j$ be an entrant at $\lambda=1$ with $w_j<x_0$. Since $\kappa(v)\to\infty$ as $v\downarrow c$, some $u>c$ has $\kappa>\Delta(j)$ on $(c,u)$, and

$$\lambda>\bar\lambda=\max\Big(1,\frac{x_0-u}{x_0-w_j}\Big)\ \Rightarrow\ x_0+\lambda(w_j-x_0)<u,$$

so rank $j$ fails at $\lambda$. By the prefix property no rank from $j$ on enters, and

$$k^*(\lambda)\le j<j+1\le k^*(1).$$

For $1\le\lambda_1\le\lambda_2$, ranks with $w_i\ge x_0$ have $i<j$ and enter at $\lambda_1$, while ranks with $w_i<x_0$ have $x_0+\lambda_2(w_i-x_0)\le x_0+\lambda_1(w_i-x_0)$, so every entrant at $\lambda_2$ enters at $\lambda_1$.

□

## Proof of M16

*Lean:*
`EntryContestPMU.weight_eq`, `EntryContestPMU.pmu_identity`, `EntryContestPMU.pmu_sign_iff`, `EntryContestPMU.kernel_rises`, `EntryContestPMU.kernel_falls`, `EntryContestPMU.kernel_max`.

With no other entrant, $H_0=CG^{Q-1}$ among $Q$ challengers and $CG^Q$ among $Q+1$, and $CG^{Q-1}-CG^Q=W_Q$, so

$$\frac{D(Q)}{V}=\int\big(CG^Q-CG^{Q-1}\big)\,dF-\int\big(CG^Q-CG^{Q-1}\big)\,dG=-\Big(\int W_Q\,dF-\int W_Q\,dG\Big).$$

For the kernel,

$$\frac{d}{dg}\,g^n(1-g)=g^{n-1}\big(n-(n+1)g\big)\quad(n\ge1),\qquad \frac{d}{dg}\,(1-g)=-1\quad(n=0).$$

□

## Proof of M17

*Lean:* `EntryContestPMU.weight_shift`, `EntryContestPMU.pmu_single_crossing`, `EntryContestPMU.pmu_step_nonpos_persists`, `EntryContestPMU.quasiconcave_of_steps`, `EntryContestPMU.pmu_quasiconcave`.

Let $k=Q_2-Q_1$, $\rho=G(x_0)^k$ and $\nu=dF-dG$. By M16 and $W_{Q_2}=G^kW_{Q_1}$,

$$D(Q_2)-\rho\,D(Q_1)=-V\int\big(G^k-\rho\big)\,W_{Q_1}\,d\nu.$$

$G$ is non-decreasing and $W_{Q_1}\ge0$, so

$$\big(G^k-\rho\big)W_{Q_1}\le0,\ \nu\le0\text{ on }(-\infty,x_0],\qquad \big(G^k-\rho\big)W_{Q_1}\ge0,\ \nu\ge0\text{ on }(x_0,\infty),$$

and $D(Q_2)\le\rho\,D(Q_1)$. Then $D(i)\le0$ gives $D(j)\le0$ for $j\ge i$. If $D(i)\le0$ for some $Q_1\le i<Q_2$, then $\Delta_{Q_3}(0)\le\Delta_{Q_2}(0)$, and otherwise $\Delta_{Q_1}(0)\le\Delta_{Q_2}(0)$.

□

## Proof of M18

*Lean:* `EntryContestPMU.uniform_left`, `EntryContestPMU.uniform_right`, `EntryContestPMU.pmu_quasiconcave_uniform`, `EntryContestRefute.pmu_step_nonneg_uniform`, `EntryContestRefute.Delta_mono_uniform`.

$G$ is the law of $\mu r$, so $G = 1$ on $[\mu,\infty)$, $dG = 0$ on $(\mu,\infty)$ and $W_Q = 0$ there. For $B \subseteq (-\infty,\mu]$ and $y \geq 0$, with $\tau = (1-\mu)y/\mu$ and $\lambda$ Lebesgue measure,

$$\Pr\big(\mu r + (1-\mu)y \in B\big) = \lambda\left([\tau,1] \cap \tfrac{B}{\mu}\right) \leq \lambda\left([0,1] \cap \tfrac{B}{\mu}\right) = \Pr(\mu r \in B),$$

so $dF \leq dG$ on $(-\infty,\mu]$ and M17 applies at $x_0 = \mu$. By M16,

$$\frac{D(Q)}{V} = \int_{(-\infty,\mu]} W_Q \, d(G-F) \geq 0.$$

□

## Proof of M19

*Lean:* `EntryContestShift.left_of_lift`, `EntryContestShift.right_of_lift`, `EntryContestShift.quasiconcave_of_shiftMono`, `EntryContestShift.shiftMono_of_density`, `EntryContestShift.lawA_shiftMono`, `EntryContestShift.witness_quasiconcave`.

Let $t = 1-\mu$ and $G_t$ be the law of $\mu r + t$, so $F = pG + (1-p)G_t$. From $\Pr(r \leq 0) = 0$, $G_t$ has no mass on $(-\infty,t]$, where $dF = p\,dG \leq dG$. For $B \subseteq (t,\infty)$, with $c = t/\mu$ and $B/\mu \subseteq (c,\infty)$,

$$G(B) = \Pr\big(r \in \tfrac{B}{\mu}\big) \leq \Pr\big(r \in \tfrac{B}{\mu} - c\big) = G_t(B),$$

so $dF \geq dG$ on $(t,\infty)$ and M17 applies at $x_0 = t$. For a density $g$ non-increasing on $(0,\infty)$ and $B \subseteq (c,\infty)$,

$$\int_B g(x)\,dx \leq \int_B g(x-c)\,dx = \int_{B-c} g(x)\,dx,$$

and the smaller of two uniform draws has $\Pr(r \leq x) = 1-(1-x)^2$ and density $2(1-x)$ on $[0,1]$.

□

## Proof of M20

*Lean:* `EntryContestFall.stepQ_one`, `EntryContestFall.stepQ_two`, `EntryContestFall.rise_then_fall`, `EntryContestFall.fall_from_one`.

Here $G(x) = 1-\big(1-\frac{4x}{3}\big)^2$ on $[0,\frac{3}{4}]$, $F(x) = pG(x) + (1-p)G(x-\frac{1}{4})$, $W_1 = F(1-G)$ and $W_2 = FG(1-G)$. By M16, integrating the piecewise polynomials on $[0,\frac{1}{4}]$, $[\frac{1}{4},\frac{3}{4}]$ and $[\frac{3}{4},1]$,

$$\frac{D(1)}{V} = -\frac{11p^2}{162} + \frac{1901p}{21870} - \frac{208}{10935}, \qquad \frac{D(2)}{V} = -\frac{4933p^2}{393660} + \frac{77219p}{2755620} - \frac{10672}{688905},$$

so

$$p = \tfrac{1}{2}: \ \frac{D(1)}{V} = \frac{653}{87480}, \ \frac{D(2)}{V} = -\frac{10169}{2204496}, \qquad p = \tfrac{1}{4}: \ \frac{D(1)}{V} = -\frac{179}{116640}.$$

□

## Proof of M21

*Lean:* `EntryContestMustFall.must_fall_general`, `EntryContestMustFall.shift_dominates`, `EntryContestMustFall.stepQ_bern`, `EntryContestMustFall.must_fall`, `EntryContestMustFall.witness_must_fall`, `EntryContestMustFall.unif_shiftMono`, `EntryContestMustFall.uniform_never_gains`.

With $F = pG + (1-p)G_t$ and $W_Q = G^{Q-1}W_1$, M16 gives

$$D(Q) = -V(1-p)\Big(\int W_Q\,dG_t - \int W_Q\,dG\Big).$$

$G_t$ has no mass on $(-\infty,t]$, $dG_t \geq dG$ on $(t,\infty)$ by M19, and $W_1 \geq \eta = C(a)\big(1-G(b)\big) > 0$ on $(a,b]$. Splitting at $t$ and $a$,

$$\int W_Q\,dG_t - \int W_Q\,dG \;\geq\; G(a)^{Q-1}A - G(t)^{Q-1}B,$$

$$A = \int_{(a,\infty)} W_1\,d(G_t - G) \geq \eta\,(G_t - G)\big((a,b]\big) > 0, \qquad B = \int_{(-\infty,t]} W_1\,dG \geq 0.$$

With $r_0 = G(t)/G(a) < 1$ and $r_0^{Q_0-1}(B+1) < A$, the bound is positive for $Q \geq Q_0$. For M20's base law, $t = \frac{1}{4}$ and

$$G\big((\tfrac{1}{2},\tfrac{5}{8}]\big) = \tfrac{35}{36} - \tfrac{8}{9} = \tfrac{1}{12} < \tfrac{7}{36} = \tfrac{3}{4} - \tfrac{5}{9} = G_t\big((\tfrac{1}{2},\tfrac{5}{8}]\big).$$

A uniform base score is shift-monotone, and M18 gives $D(Q) \geq 0$, so it never meets the condition.

□

## Proof of M22

*Lean:* `EntryContestPMUWitness.pairA_fosd`, `EntryContestPMUWitness.pairB_fosd`, `EntryContestPMUWitness.pairA_step_one`, `EntryContestPMUWitness.pairA_step_two`, `EntryContestPMUWitness.pairB_step`, `EntryContestPMUWitness.fosd_does_not_sign`.

Pair one, with $y = 1-x$, $W_1 = x(1-x)^2$, $W_2 = x\big(1-(1-x)^2\big)(1-x)^2$ and $dG = 2(1-x)\,dx$, gives

$$\int W_1\,dF = \tfrac{1}{12}, \quad \int W_1\,dG = \tfrac{1}{10}, \quad \int W_2\,dF = \int_0^1 (y^2 - y^3 - y^4 + y^5)\,dy = \tfrac{1}{20}, \quad \int W_2\,dG = \tfrac{11}{210},$$

$$\frac{D(1)}{V} = \tfrac{1}{10} - \tfrac{1}{12} = \tfrac{1}{60}, \qquad \frac{D(2)}{V} = \tfrac{11}{210} - \tfrac{1}{20} = \tfrac{1}{420}.$$

Pair two, with $W_Q = x^{Q+1} - x^{Q+2}$ and $dF = 2x\,dx$, gives

$$\int W_Q\,dF = \frac{2}{(Q+3)(Q+4)}, \qquad \int W_Q\,dG = \frac{1}{(Q+2)(Q+3)}, \qquad \frac{D(Q)}{V} = -\frac{2(Q+2)-(Q+4)}{(Q+2)(Q+3)(Q+4)}.$$

□

## Proof of M23

*Lean:* `EntryContestKappa.burden_strictAnti_of_strictConcave`, `EntryContestBM.burden_strictAnti_of_deriv`, `EntryContestBM.not_burden_of_convex_stretch`, `EntryContestKappa.log_burden_strictAnti`, `EntryContestKappa.crra_strictConcave`, `EntryContestKappa.crra_burden_strictAnti`.

For $u$ strictly concave and $c < a < b$, secant slopes fall as either end moves right, and $a-c < a < b$, $a-c < b-c < b$ give

$$\frac{u(a)-u(a-c)}{c} > \frac{u(b)-u(a-c)}{b-a+c} > \frac{u(b)-u(b-c)}{c},$$

so $\kappa(a) > \kappa(b)$. With $u$ differentiable,

$$\kappa'(w) = u'(w) - u'(w-c) < 0 \quad (w > c).$$

If $u$ is convex on $[a,b]$ with $b-a>c$, then $a<b-c$ and

$$\kappa(a+c)=u(a+c)-u(a)\le u(b)-u(b-c)=\kappa(b).$$

Log and $x^{1-\gamma}/(1-\gamma)$ have $u''<0$ on $(0,\infty)$.

□

## Proof of M24

*Lean:* `EntryContestBM.rampU_burden`, `EntryContestBM.rampU_kappa_diverges`, `EntryContestBM.rampU_strictConvex`, `EntryContestBM.rampU_not_concave`, `EntryContestBM.admissibleParams_open`, `EntryContestBM.rampU_admissible`, `EntryContestBM.admissible_width`, `EntryContestBM.bm_strictly_weaker`.

Let $\rho$ rise linearly from 0 at $x_0$ to 1 at $x_0+\sigma$, and

$$u(x)=\log x+x+\eta\int_0^x\rho(t)\,dt,\qquad u'(x)=\tfrac{1}{x}+1+\eta\rho(x)>0.$$

On $[x_0,x_0+\sigma]$, $u''=-x^{-2}+\eta/\sigma>0$ when $\sigma<\eta x_0^2$, so $u$ is strictly convex there and not concave. For $w>c$, $\rho(w)-\rho(w-c)\in[0,1]$ and vanishes unless $x_0<w<x_0+\sigma+c$, where

$$\tfrac{1}{w-c}-\tfrac{1}{w}=\tfrac{c}{w(w-c)}>\tfrac{c}{(x_0+\sigma+c)(x_0+\sigma)}>\eta,$$

so $u'(w)<u'(w-c)$ and M23 applies. $u(x)\to-\infty$ as $x\downarrow 0$, so M25 gives the divergence. The conditions

$$x_0>0,\quad \sigma>0,\quad \sigma<\eta x_0^2,\quad \eta(x_0+\sigma+c)(x_0+\sigma)<c$$

are strict inequalities in continuous functions, so they define an open set, and for $\sigma<c$ an $\eta$ between $\sigma/x_0^2$ and $c/\big((x_0+\sigma+c)(x_0+\sigma)\big)$ exists once $x_0$ is large, since $\sigma(x_0+\sigma+c)(x_0+\sigma)/x_0^2\to\sigma<c$.

□

## Proof of M25

*Lean:* `EntryContestKappa.kappa_diverges_iff`, `EntryContestKappa.not_diverges_of_continuousWithinAt_zero`, `EntryContestKappa.log_kappa_diverges`, `EntryContestKappa.crra_kappa_diverges`, `EntryContestKappa.crra_kappa_not_diverges`.

$u(w)\to u(c)$ as $w\downarrow c$, so

$$\kappa(w)=u(w)-u(w-c)\to\infty\iff u(w-c)\to-\infty\iff u(x)\to-\infty\ (x\downarrow 0).$$

$\log x\to-\infty$, and

$$\frac{x^{1-\gamma}}{1-\gamma}\to-\infty\quad(\gamma>1),\qquad \frac{x^{1-\gamma}}{1-\gamma}\to 0\quad(\gamma<1).$$

□

## Proof of M26

*Lean:* `EntryContestAnon.profile_mem`, `EntryContestAnon.profile_not_mem`, `EntryContestAnon.cdf_profile`, `EntryContestAnon.gain_anonymous`.

Challenger $i$ wins with probability $\int C\prod_{j\ne i}L_j\,dL_i$, where $L_j$ is $F$ for an entrant and $G$ for an outsider. With $|T|=m$ and $i\notin T$,

$$\frac{\mathrm{gain}_i(T)}{V}=\int CF^mG^{Q-1-m}\,dF-\int CF^mG^{Q-1-m}\,dG=\int H_m\,dF-\int H_m\,dG=\frac{\Delta(m)}{V}.$$

□

## Proof of M27

*Lean:* `EntryContestAnon.winProb_zero_entrant`, `EntryContestAnon.winProb_zero_outsider`, `EntryContestAnon.gain_zero`, `EntryContestAnon.kappa3_values`, `EntryContestAnon.gains3`, `EntryContestAnon.anon_witness_zero`, `EntryContestAnon.gain_tendsto`, `EntryContestAnon.anon_witness_positive`, `EntryContestAnon.anon_witness_exists`.

Let entrant $j$'s $s_j$ be the largest of $n_j$ uniform draws, the incumbent's the largest of one, and $\mu = 0$. An outsider scores 0, and among the $1+\sum_{j\in S} n_j$ draws each is equally likely to be the largest, so

$$\Pr(i \text{ wins}) = \frac{n_i}{1+\sum_{j\in S} n_j}, \qquad \operatorname{gain}_i(T) = \frac{V\, n_i}{1+n_i+\sum_{j\in T} n_j}.$$

With $V = 1$, $c = 1$, $u(x) = -1/x$, $w = (\frac{11}{4}, \frac{9}{4}, 2)$ and $n = (1,2,3)$,

$$\kappa = \left(\tfrac{16}{77}, \tfrac{16}{45}, \tfrac{1}{2}\right), \qquad \operatorname{gain}_2(\emptyset) = \tfrac{3}{4},\ \operatorname{gain}_0(\{2\}) = \tfrac{1}{5},\ \operatorname{gain}_1(\{2\}) = \tfrac{1}{3},$$

$$\operatorname{gain}_0(\{1\}) = \tfrac{1}{4},\ \operatorname{gain}_1(\{0\}) = \tfrac{1}{2},\ \operatorname{gain}_2(\{0,1\}) = \tfrac{3}{7},$$

so $\{2\}$ and $\{0,1\}$ are equilibria. Every inequality is strict and each gain is continuous in $\mu$ at 0, so both remain equilibria for small $\mu > 0$.

□

## Proof of M28

*Lean:* `EntryContestMu.Delta_normalise`, `EntryContestMu.cdf_payerLaw_antitone`, `EntryContestMu.first_integral`, `EntryContestMu.Delta_payer_mono`, `EntryContestMu.Delta_antitone_mu`, `EntryContestMu.kstar_antitone_mu`.

For $\mu > 0$ put $t = (1-\mu)/\mu$, $F_t$ the law of $r + ts$ and $G$ the law of $r$. Dividing every score by $\mu$ changes no rank, so with $C = F$ and $n = Q - 1 - m$,

$$\Delta(m;\mu) = V\Big(\int F_t^{\,m+1} G^n \, dF_t - \int F_t^{\,m+1} G^n \, dG\Big).$$

For $s \ge 0$ and $t \le t'$, $r + t's \ge r + ts$, so $F_{t'}(x) \le F_t(x)$ for every $x$, and the second integral is non-increasing in $t$. For the first, with $n \ge 1$,

$$\int F_t^{\,m+1} G^n \, dF_t = \frac{1}{m+2} \int G^n \, d\big(F_t^{\,m+2}\big) = \frac{1}{m+2}\Big(1 - \int F_t^{\,m+2} \, d\big(G^n\big)\Big),$$

by $\int g F^k \, dF = \frac{1}{k+1}\int g\, d(F^{k+1})$ and $\int A\, dB + \int B\, dA = 1$ for atomless $B$; the last integral is non-increasing in $t$. For $n = 0$ the first integral is $1/(m+2)$. So $\Delta(m;\mu)$ is non-decreasing in $t$, and $t$ is decreasing in $\mu$. With $\kappa$ fixed, if $k'^* < k^*$ then

$$\Delta(k'^*;\mu') < \kappa_{k'^*} \le \Delta(k'^*;\mu) \le \Delta(k'^*;\mu'),$$

so $k^*(\mu) \le k^*(\mu')$ for $\mu' \le \mu$.

□

## Proof of M29

*Lean:* `EntryContestAligned.kappaA_values`, `EntryContestAligned.ability_with_wealth`, `EntryContestAligned.costs_ordered`, `EntryContestAligned.gainsA`, `EntryContestAligned.aligned_witness_zero`, `EntryContestAligned.aligned_witness_positive`, `EntryContestAligned.aligned_witness_exists`, `EntryContestAligned.control_window`.

Let $s_j$ be the largest of $n_j$ uniform draws with $n = (12, 10, 9)$, the incumbent's the largest of one, and $\mu = 0$. An outsider scores 0 and never wins, and for $i \in S$,

$$\operatorname{gain}_i(S \setminus \{i\}) = \frac{n_i}{n_i + 1 + \sum_{j\in S\setminus\{i\}} n_j},$$

so

$$\mathrm{gain}_0(\emptyset) = \tfrac{12}{13},\ \mathrm{gain}_0(\{1,2\}) = \tfrac{3}{8},\ \mathrm{gain}_1(\{0\}) = \tfrac{10}{23},\ \mathrm{gain}_1(\{2\}) = \tfrac{1}{2},\ \mathrm{gain}_2(\{0\}) = \tfrac{9}{22},\ \mathrm{gain}_2(\{1\}) = \tfrac{9}{20}.$$

With $u(x) = -1/x$, $c = 1$ and $w = (\frac{43}{20}, \frac{52}{25}, \frac{83}{40})$, $\kappa = (\frac{400}{989}, \frac{625}{1404}, \frac{1600}{3569})$, increasing in the index, and

$$\tfrac{3}{8} < \kappa_0 \le \tfrac{12}{13}, \qquad \tfrac{10}{23} < \kappa_1 \le \tfrac{1}{2}, \qquad \tfrac{9}{22} < \kappa_2 \le \tfrac{9}{20},$$

so $\{0\}$ and $\{1,2\}$ are equilibria. Every inequality is strict and each gain is continuous in $\mu$ at 0, so both remain equilibria for small $\mu > 0$. Raising $\kappa_2$ to $\frac{23}{50} > \frac{9}{20}$ removes $\{1,2\}$.

□

## Proof of M30

*Lean:* `EntryContestInc.Delta_incumbent_in_eq`, `EntryContestInc.symGain_step`, `EntryContestInc.incumbent_out_iff`, `EntryContestInc.incumbent_out_not_unique`, `EntryContestInc.incumbent_in_of_pinned`, `EntryContestInc.band_witness`, `EntryContestInc.control_no_band`.

Players $0, \dots, Q$, costs $\kappa_0 \le \cdots \le \kappa_Q$, gain $g(m)$ with $m$ others in. For $m+2 \le Q$,

$$F \cdot F^m G^{Q-1-m} = G \cdot F^{m+1} G^{Q-2-m},$$

so $\Delta(m)|_{C=F} = g(m+1)$, and $g(m+1) \le g(m)$ by M4. Let $S$ be an equilibrium with $0 \notin S \neq \emptyset$. By M6, $|S| = k^* \ge 1$; $S \subseteq \{1, \dots, Q\}$ gives $k^* \le Q$; $0 \notin S$ gives $g(k^*) < \kappa_0$; and $|\{1, \dots, k^*-1\}| = k^* - 1 < |S|$ gives $i \in S$ with $i \ge k^*$, so

$$\kappa_{k^*} \le \kappa_i \le g(k^* - 1).$$

Conversely, let $S = \{1, \dots, k^*\}$. For $i \in S$, $\kappa_i \le \kappa_{k^*} \le g(k^*-1) = g(|S|-1)$, and for $j \notin S$, $g(|S|) = g(k^*) < \kappa_0 \le \kappa_j$. If $g(k^*-1) < \kappa_{k^*}$ the condition fails, so $0 \in S$ for every non-empty equilibrium $S$; $\{0, \dots, k^*-1\}$ is an equilibrium by M8. For $\kappa = (2,5,8)$ and $g = (12,10,1)$, $k^* = 2$, $g(2) = 1 < 2$, $\kappa_2 = 8 \le 10 = g(1)$, and $\{1,2\}$ is an equilibrium; with $\kappa_2 = 11 > 10$ no equilibrium omits 0.

□

## Proof of M31

*Lean:* `EntryContestShare.payerShare_full`, `EntryContestShare.payerShare_eq_one_sub`, `EntryContestShare.payerShare_monotone`, `EntryContestShare.payerShare_at_equal`, `EntryContestShare.payerShare_normalise`, `EntryContestShare.payerShare_antitone_mu`, `EntryContestShare.payerShare_kstar_antitone_mu`, `EntryContestShare.control_share_needs_nonneg`.

The best of $k+1$ payers has law $F^{k+1}$ and the best of $n = Q-k$ outsiders $G^n$, so $\xi(k) = \Pr(\text{best payer} > \text{best outsider})$, and $\xi(Q) = 1$. For $n \ge 1$, by $\int A\,dB + \int B\,dA = 1$ for atomless $B$,

$$\xi(k) = 1 - \int F^{k+1}\, d(G^n),$$

$$\xi(k+1) - \xi(k) = \int F^{k+1}\, d(G^n) - \int F^{k+2}\, d(G^{n-1}) \ge 0,$$

since $F^{k+2} \le F^{k+1}$ and $\int F^{k+1}\, d(G^{n-1}) \le \int F^{k+1}\, d(G^n)$ by the same identity and $G^n \le G^{n-1}$. For $F = G$,

$$\xi(k) = (k+1) \int F^{Q-k} F^k\, dF = \frac{k+1}{Q+1}.$$

For $\mu > 0$, dividing every score by $\mu$ changes no rank, so $\xi$ is computed from $F_t$ and $G$ with $t = (1-\mu)/\mu$; $F_{t'} \le F_t$ for $t \le t'$ (M28), so $\xi(k)$ is non-decreasing in $t$ and non-increasing in $\mu$. With $\kappa$ fixed and $\mu' \le \mu$, $k^*(\mu) \le k^*(\mu')$ by M28, and

$$\xi\big(k^*(\mu); \mu\big) \le \xi\big(k^*(\mu); \mu'\big) \le \xi\big(k^*(\mu'); \mu'\big).$$

With $s = -1$, $r$ uniform, $Q = 1$ and $k = 0$, $\xi(0) = \frac{1}{2}$ at $t = 0$ and 0 at $t = 1$, so $s \ge 0$ is needed.

□

# B Evidence and verification

Every row of the model table is a theorem of Lean 4 with Mathlib. The Lean column names the theorems, and the proof of each row in Appendix A names the same ones. The development lives in `lean/mathlib/`, built on the toolchain `v4.24.0-rc1` against Mathlib commit `14871d5`, beside the core file `lean/EntryContest.lean`, which uses no Mathlib. The script `verify.sh` runs three kinds of check on every build, and `VERIFICATION.md` records what they printed.

1. *The axiom audit.* No source file contains `sorry`. Every declared theorem is passed to `#print axioms`. A Mathlib theorem that depends on an axiom other than `propext`, `Classical.choice` and `Quot.sound` fails the build, and a theorem of the core file that depends on an axiom other than `propext` and `Quot.sound` fails it too.

2. *The manuscript checker* (`checks/verify_manuscript.py`). Every model row names audited theorems and has exactly one proof naming the same theorems. Introduction rows point to model rows, and conclusion rows point to model or literature rows. Every literature quote appears verbatim in the cited paper's record under `lit/`. Plain English is at most 30% of every proof. Each rule also runs on a faulty sample, which the rule must reject.

3. *The literature records.* Each cited paper has a directory under `lit/` holding a reconstruction from the source, the claims our documents make about the paper with page locators, notes on every discrepancy found, and Lean for the mathematical claims. A record checks our reading of a source and does not reprove the source. The records run under `lit/verify_lit.sh`.

**What Lean does not carry.** The standing assumptions of Appendix D enter the theorems as hypotheses. The rows assume that $F$ and $G$ have no atoms rather than deriving it. The assumption follows from the primitives when $r$ has no atoms and $\mu \neq 0$ (`EntryContestRefute.investorLaw_noAtoms'`, `EntryContestRefute.nonInvestorLaw_noAtoms'`). A statement about what a cited paper says rests on a quotation with its page, recorded under `lit/`, and not on Lean.

**Illustrations.** The SymPy and numerical scripts under `checks/` still run in `verify.sh`, as illustrations and as regression tests. No row rests on them. Earlier drafts graded each claim by an evidence tier (Lean, SymPy or numerical) because several analytic steps entered the Lean development only as hypotheses. Those steps are now proved in Lean, and the tiers are retired.

# C Terminology

Every English term that carries load in a row or a proof is bound here to a symbol built from the primitives of Appendix D. Each entry names an object already defined, so the table is a translation key and not a second set of definitions. Indices follow Appendix A, with the richest challenger at index 0.

| Term | Symbol | Meaning |
|---|---|---|
| *Players and actions* | | |
| challenger | $i \in \{0,\dots,Q-1\}$ | A player of given wealth who chooses whether to compete. |
| incumbent | $C \in \{F,G\}$ | The additional player, who competes in every case and appears only through her score CDF, which is $F$ when she holds the bought component, as M9, M20, M22, M28 and M31 take, and $G$ when she does not. |
| enter | $e_i = 1$ | Pay the fee $c$ and add $(1-\mu)s_i$ to the score. |
| entrant, outsider | $e_i = 1$, $e_i = 0$ | A challenger who enters, and one who does not. |
| entrant set | $S = \{i : e_i = 1\}$ | The challengers who enter. |
| prize | $V > 0$ | Fixed and awarded by rank, outside $u$. |
| *Wealth and cost* | | |
| wealth | $w_i$ | Challenger $i$'s resource. A challenger with $w_i \leq c$ does not enter. |
| wealth profile | $(w_0,\dots,w_{Q-1})$ | The realised wealths, common knowledge, sorted with $w_0 \geq w_1 \geq \cdots$. |

| Term | Symbol | Meaning |
|---|---|---|
| entry cost | $\kappa(w) = u(w) - u(w-c)$ | The burden of the fee in utility units. |
| fee | $c$ | The nominal cost of entry, the same for every challenger. |
| burden-monotonicity | $\kappa$ strictly decreasing on $(c, \infty)$ | The assumption on $u$ (M23), weaker than strict concavity (M24). |
| *Scores and gains* | | |
| score | $\sigma_i = \mu r_i + (1-\mu) s_i e_i$ | The prize goes to the highest score. |
| score CDFs | $F$, $G$ | The laws of an entrant's score and an outsider's, with $F \leq G$ (M2). |
| dominance gap | $\varphi = G - F \geq 0$ | What entry buys. |
| best rival score | $H_m = CF^m G^{Q-1-m}$ | The CDF of the best score facing a challenger when $m$ others enter. |
| gain | $\Delta(m)$ | The value of entering when $m$ others enter, with $\Delta(m+1) \leq \Delta(m)$ (M4). |
| anonymity | $\Delta$ depends on $m$ alone | The gain depends on how many others enter and not on who they are (M26, M27). |
| first entrant's gain | $\Delta_Q(0)$, $D(Q)$ | The gain with no other entrant among $Q$ challengers, and its step in $Q$ (M16 to M22). |
| *Equilibrium* | | |
| entry condition | $\kappa_j \leq \Delta(j)$ | Index $j$ enters when its cost is at most its gain. |
| entry count | $k^*$ | The number of indices before the first one whose entry condition fails (M5). |
| count invariance | $\lvert S \rvert = k^*$ | Every pure-strategy equilibrium has $k^*$ entrants (M6). |
| identity multiplicity | $S \neq S'$, $\lvert S \rvert = \lvert S' \rvert = k^*$ | Different entrant sets can be equilibria at the same count (Appendix D). |
| assortative set | $\{0, \ldots, k^* - 1\}$ | The $k^*$ richest challengers (M7, M8). |
| marginal entrant | index $k^* - 1$ | The poorest entrant of the assortative set. |
| first outsider | index $k^*$ | The richest challenger outside the assortative set. |
| *Comparative statics* | | |
| change of profile | $w \mapsto w'$ | Any map between sorted wealth profiles. No result assumes it keeps the mean unless it says so. |
| mean-preserving spread | $\sum_j w'_j = \sum_j w_j$, $w'$ majorizing $w$ | The case of economic interest, and not a hypothesis of M12. |
| displacement | $d_j = w'_j - w_j$ | The rank-by-rank difference of the two sorted profiles. |
| pivot-spread | $T$ with $T(w) \geq w$ for $w \geq x_0$ and $T(w) \leq w$ for $w \leq x_0$ | A non-decreasing map that spreads wealth on both sides of the pivot $x_0$ (M13). |
| ordering in dispersion | $z - y$ non-decreasing | For quantile functions $y$ and $z$, the second law is more dispersed [Lewis and Thompson, 1981]. With a crossing point, the transport map between the two laws is a pivot-spread. |
| margin condition | $d_{k^*-1} \geq 0$, or $d_{k^*} \leq 0$ | The hypothesis of the rise branch, read at the marginal entrant, and of the fall branch, read at the first outsider (M12). |
| band | $d_{k^*-1} < 0 < d_{k^*}$ | The margins on which the margin condition is silent (M14). |
| *Aggregates* | | |
| paying class's share | $\xi(k^*)$ | The probability that the prize goes to one who paid, the incumbent included (M31). |
| total entry cost | $\sum_{i \in S} \kappa(w_i)$ | In utility units. Its resource counterpart $\lvert S \rvert c = k^* c$ is the same in every equilibrium. |
| sum of payoffs | $\sum_i U_i$ | Unweighted, so an interpersonal comparison of the index $u$ (M8). |
| selection | a rule picking one equilibrium | Not a refinement. Nothing claims that players coordinate on it. |

Three pairs are kept apart because earlier drafts conflated them. The entry cost in utility, $\kappa$, varies with wealth, while the fee in resources, $c$, does not and is what society gives up. Count invariance holds, while identity multiplicity is real, and every comparative static concerns the count for that reason. A change of profile is what the results apply to, while the margin condition is what they assume.

# D Primitives and scope

**The contest.** One prize of value $V > 0$ is fixed and exogenous, and it is awarded by rank rather than sold. $Q$ challengers and one incumbent compete. Each challenger chooses $e_i \in \{0, 1\}$, and entry costs the fee $c$. The draws $r_i$ and $s_i \geq 0$ are independent across players and independent of the wealth profile, from arbitrary laws. The score is $\sigma_i = \mu r_i + (1-\mu) s_i e_i$, the highest score takes the prize, and the game is static and played once. By M1 the gain is proportional to $V$, so $V$ enters only as a scale. Cole et al. [1992] read status as "a ranking device that determines how well he or she fares in the nonmarket sector", which is the sense of a prize awarded by rank.

**The incumbent.** The incumbent is scored by the rule that admits the challengers. Her law is $C = F$, so she holds the bought component and differs from an entrant in nothing, and M26 includes her. This is the open contest, in which the holder competes again under the same rule. Her entry is a parameter in M1 to M29. In M30 it is a choice, she is player 0 of a game of $Q+1$ players with the lowest cost, and outside the band of M6 she is in every non-empty equilibrium, so $C = F$ is the equilibrium outcome there. Two institutions are not modelled, and this choice turns them off. A protected holder, who keeps the position unless the best challenger clears a fixed bar, is a law $C$ that stays put as $\mu$ varies. A holder with an edge beyond wealth is a law above $F$, and sits at the boundary of M27. Every row holds for an arbitrary atomless $C$ except M9, M20, M22, M28 and M31, which take $C = F$, M29, whose witness fixes $C$, M30, where her entry is chosen, and M21, which needs $C(a) > 0$. A fixed $C$ therefore keeps the count, the sorting, the cap and the spread results, and loses the two $\mu$ benchmarks and the directions in $\mu$ of M28 and M31.

**Payoffs.** With $u$ the utility of wealth,

$$U_i = u(w_i - c\,e_i) + V\,\mathbf{1}\{i \text{ wins}\}.$$

The fee is paid inside $u$, so its burden is felt in utility and varies with wealth. The prize enters outside $u$ and linearly, so rich and poor value it alike. Inequality therefore acts through one channel, the cost of entry. This is the privilege contest of Schroyen and Treich [2016], and not their rent-seeking contest, in which a money prize would sit inside $u$.

**After entry.** The model stops at entry. One prize is awarded by rank, and nothing after the award is modelled. Differences in pay among those who hold positions, such as the class pay gap of Laurison and Friedman [2016], are outside it; the comparable quantity is the share of positions held by those who paid (M31).

**No effort margin.** Entry costs a fixed fee and buys a fixed component of the score, and there is no effort to choose. A more accurate ranking of that component therefore only raises what paying buys, and the count does not fall as the component weighs more (M28). With an effort choice the sign is the opposite (L20, L22). In Morgan et al. [2022] a more accurate ranking raises the marginal return to effort, effort rises until competing earns nothing, and the prizes then clear only if some contestants drop out. With logistic noise of scale $\sigma$, cost $k\,e^{\alpha x}$ in log-output and a mass $m$ of prizes, their equilibrium conditions give participation $m/(1-\sigma\alpha)$ in the drop-out region, because the logistic hazard equals the distribution function. Participation rises with the noise, from $m$ as $\sigma \to 0$ to 1 at $\sigma = (1-m)/\alpha$, and their second-order condition holds throughout (`MeritocracyLogistic.mixed_region`, `MeritocracyLogistic.participation_strictMonoOn`, `MeritocracyLogistic.soc_mixed`). This instance is derived here in their model, and they do not state it. In the free-entry extension of Drugov and Ryvkin [2020] the less dispersed noise brings higher effort and weakly fewer entrants, the same sign for the same reason (L22).

**Utility.** $u$ is strictly increasing and burden-monotone, so $\kappa(w) = u(w) - u(w-c)$ is strictly decreasing on $(c,\infty)$. Strict concavity implies it (M23), and it is strictly weaker (M24). One $u$ serves every challenger, so $\kappa$ varies across challengers through wealth alone. $u$ is not observed. It represents the recorded entries rather than determining them, preference being "the binary representation of individual choice" [Sen, 1973, p. 253] and utility being "used to represent preferences" [Schoemaker, 1982, p. 532], and it enters the model only through $\kappa$. $\kappa$ is a difference of such a $u$ and not a measure of how strongly the fee is felt [Schoemaker, 1982, p. 533]. The statement that $\kappa$ falls with wealth therefore has no content of its own. Its content is what it implies for a record of wealths, fee, prize and entries. The richest $k^*$ challengers entering is an equilibrium, and the only one when $\Delta(k^*-1) < \kappa_{k^*}$ (M7, M8), and the count moves with the prize, the fee and the profile as M11 to M13 state. A challenger whose wealth is at or below $c$ does not enter. The strictness result M15 also uses $\kappa(w) \to \infty$ as $w \downarrow c$, which holds exactly when $u(x) \to -\infty$ as $x \downarrow 0$ (M25).

**Score laws.** $F$ is the law of $\mu r + (1-\mu)s$ and $G$ the law of $\mu r$, and both are taken to have no atoms (Appendix B). Earlier drafts also assumed a common lower endpoint for $F$ and $G$ and that $\varphi$ vanishes at both ends of the support. No row uses either, and both are dropped.

**Information.** The realised wealth profile is common knowledge, entry is simultaneous, and the draws $r_i$ and $s_i$ are made after entry. At the entry stage a challenger knows $c$, $V$, $\mu$, the laws of $r$ and $s$ and the wealth profile, and not her own draws, which reach her decision only through $F$ and $G$. Her condition $\kappa(w_i) \le \Delta(j)$ has $w_i$ as its only individual datum, so talent enters neither the cost of entry nor the decision. Because the profile is common knowledge, the entry condition is written on the sorted profile, and $k^*$ is a count shared by every pure-strategy equilibrium (M6). It does not pin who enters. With $\kappa = (2,5,8)$ and $\Delta = (12,10,1)$, the sets $\{0,1\}$, $\{0,2\}$ and $\{1,2\}$ are all equilibria, the last without the richest challenger (`EntryContestEq.three_equilibria`). Identities are pinned when $\Delta(k^*-1) < \kappa_{k^*}$ (M7). Under private entry costs the count would be binomial and the entrant set random, as in Moreno and Wooders [2011], where every buyer compares her own cost with one threshold. Here the threshold depends on the index, $\kappa_j \le \Delta(j)$, and it is flat only when $\Delta$ is constant, which M4 rules out whenever $\int \varphi^2 \, dK_m > 0$.

**The anonymity boundary.** Independence of the draws from wealth makes $F$ and $G$, and so $\Delta$, depend on the count of entrants alone (M26). Economies in which wealth buys ability are outside every row except M27 and M29, which mark the boundary. The exclusion is shown and not only asserted. When ability falls with wealth, equilibria of different sizes coexist (M27), and count invariance fails. It also fails for some costs when ability rises with wealth (M29). The question of how many enter and the question of which enter then cease to be separable, and every comparative static of the model concerns the count.

# E Refuted conjectures

**R1, a collapse threshold.** An earlier draft conjectured a finite $\bar{Q}$ beyond which $k^* = 0$. The conjecture is false. With $r$ uniform on $[0,1]$, $s = 0$ with probability $p$ and $s = 1$ otherwise, $0 < \mu \le \frac{1}{2}$ and an investing incumbent, for every $Q \ge 1$

$$\Delta_Q(0) = V\Big(\frac{1-p^2}{2} - \frac{p(1-p)}{Q+1}\Big) \ge V\,\frac{1-p}{2}, \qquad \Delta_Q(0) \to V\,\frac{1-p^2}{2},$$

so a challenger whose cost is at most $V(1-p)/2$ enters for every $Q$ (`EntryContestRefute.bern_Delta`, `EntryContestRefute.bern_Delta_lower`, `EntryContestRefute.bern_Delta_limit`). When an outsider's score never exceeds $x_G$, $\Delta(m) \ge V\big(\int_{x \ge x_G} C F^m \, dF - \frac{1}{Q-m}\big)$ for $m < Q$, and the first term does not depend on $Q$ (`EntryContestRefute.r1_lower_bound`). An earlier draft gave the limit as $\Pr(X > \mu)$ times the chance of beating the incumbent. That formula is withdrawn, since in the family above $\Pr(X > \mu) = 1 - p$ (`EntryContestRefute.bern_investor_above`) while the limit is $V(1-p)(1+p)/2$.

**R2, fewer entrants as $Q$ grows.** A referee conjectured that for $\mu > 0$ a larger $Q$ shrinks the gain, so fewer enter. The conjecture is not general. In the family of R1 with $0 < p < 1$, $\Delta_Q(0)$ rises strictly at every $Q$ (`EntryContestRefute.bern_Delta_strictMono`), so a challenger whose cost lies between $\Delta_Q(0)$ and $\Delta_{Q+1}(0)$ stays out with $Q$ challengers and enters with $Q+1$. With $r$ uniform the gain never falls in $Q$ (M18). The conjectured direction occurs for other primitives (M20), and the sign of the effect is set by the law of $r$ (M16 to M22).

# F Responses to the referee reports

**Asymmetric $k$-of-$Q$ equilibria.** Both referees found that the earlier version compared aggregate participation regimes instead of checking unilateral deviations. The model now derives the asymmetric equilibrium in which $k^*$ of the $Q$ challengers enter, checks deviations index by index, and shows the count is the same in every equilibrium (M5 to M8).

**Prize and affordability.** The second referee objected that the earlier version used one quantity both as inequality and as the prize, which made the inequality result close to tautological, and that a larger prize should raise every participant's incentive. The prize is now fixed and separate from the wealth distribution (Appendix D). A larger prize raises every gain and does not lower the entry count, and a larger fee does not raise it (M11). The inequality result is the sign rule of M12 to M15, in which $V$ does not move.

**The mobility ceiling.** Both referees found the old ceiling an artefact of uniform supports and of a limit in $Q$ under which the equilibrium does not exist. The saturation bound replaces it. An entrant at index $m$ with cost $\kappa$ has $(m+1)\kappa \le V$, so the marginal entrant gives $k^* \le V/\kappa_{k^*-1}$ for every $Q$, with no limit taken (M10).

# G Coverage of the referee comments

The two reports on the earlier submission (BEJEAP, rejected 31 August 2026) are recorded comment by comment in `HANDOVER_referee_reports_20260831.md`. Each comment is marked *answered* when a model row or an appendix settles it, *partly answered* when part of it remains, *open* when nothing here settles it, and *moot* when it concerned text or an object that the rebuilt model no longer has. Comments that only praised a change are omitted.

| Comment | Substance | Status | Where |
|---|---|---|---|
| *Reviewer 1* | | | |
| R1.1 | Aggregate regimes were compared in place of unilateral deviations. | answered | Deviations are checked index by index. The count $k^*$ is the same in every pure-strategy equilibrium, and the assortative set is one (M5 to M8). |
| R1.2 | The reach into education policy is limited by a zero-sum, fixed-slot contest. | answered | The rebuilt model makes no policy claim. The prize is fixed and awarded by rank, and the scope is stated with it (Appendix D). |
| R1.3 | Results were asserted beyond uniform distributions without proof. | answered | Every row holds for general laws of $r$ and $s$ unless it names a family or a class, and the rows that do say so (M18 to M22, M27). |
| R1.4 | The contribution is not located against contests with entry costs, positional education and status competition. | partly answered | The literature table (L1 to L19) places each result. L14 to L16 place the status reading against Weber's distinction of class from status, and L17 to L19 against access to positions, closure and opportunity hoarding (C10). The novelty reading of the remaining papers is pending, and Table 5 says what each novelty claim waits on. |
| R1.4 | The mobility ceiling reflects the uniform specification. | answered | The ceiling is replaced by a bound for every $Q$ and every law, with no limit taken (M10). |
| R1.a2 | "Stable Nash equilibrium" is undefined. | answered | Only pure-strategy Nash equilibrium is used (M6). |
| R1.a3 | Full withdrawal should be a third regime. | answered | The count $k^*$ ranges over $0,\dots,Q$, so no entry, partial entry and full entry are all cases of one result (M5, M6). |
| R1.a4 | The ceiling is used inconsistently across regimes. | answered | One bound covers every count (M10). |
| R1.a5 | Prize effects and affordability effects are conflated. | answered | The prize sits outside $u$ and the fee inside it (Appendix D). The prize and the fee move the count separately (M11). |
| R1.a6 | The notation $\Phi(Q)$ is inconsistent. | moot | The rebuilt model has no $\Phi$. |
| R1.a7 | Comparative statics in $\mu$ were misstated. | open | No comparative static in $\mu$ is claimed. |
| R1.a8 | The paper is verbose. | answered | Results sit in tables, and every proof is at most 30% prose, checked by script (Appendix B). |
| *Reviewer 2* | | | |
| R2.1 | Asymmetric equilibria in which only some challengers enter. | answered | The equilibrium is the count $k^*$ of entrants, and every equilibrium has it (M5, M6). |
| R2.1 | More enter as the prize rises and as the fee falls. | answered | M11. |
| R2.1 | More enter as social capital matters more ($\mu$ falls). | answered | With the incumbent investing, the gain and the count are non-increasing in $\mu$ for every law of $r$ and of $s \ge 0$ (M28). |
| R2.1 | At $\mu = 0$ the number entering does not depend on $Q$. | answered | At $\mu = 0$ the gain is $V/(m+2)$ for every $Q$ (M9), so with the same costs the count is the same. |
| R2.1 | At $\mu > 0$ a larger $Q$ means fewer enter. | answered | Not in general. The first entrant's gain can rise in $Q$, never falls with a uniform base score, and the sign depends on the law of $r$ (M16 to M22, conjecture R2 in Appendix E). |
| R2.1 | An equilibrium in which the incumbent abstains and a challenger enters. | answered | Only inside the band. With the incumbent as player 0 of a game of $Q+1$ players, such an equilibrium exists iff $g(k^*) < \kappa_0$ and $\kappa_{k^*} \le g(k^*-1)$, and the assortative set, with her in, is an equilibrium in every case (M30). |

| Comment | Substance | Status | Where |
|---|---|---|---|
| R2.2 | The effect of inequality on status expenditure is unclear, and the prize is the inequality. | answered | The prize is fixed and apart from the wealth distribution. A spread changes the count, and so the resource expenditure $k^*c$, with a sign set by the marginal entrant against the pivot (M12 to M15). |
| R2.2 | Endowment inequality should reduce status expenditure, as in the literature. | answered | It does when the marginal entrant is at or below the pivot, and the reverse holds when it is above (M13). |
| R2.3 | Policy is discussed without an objective. | partly answered | No policy claim is made. Among equilibria of one size the assortative set has the least total utility cost, and the resource cost $k^*c$ is the same in all of them (M6, M8). No welfare section is attempted. |
| R2.4 | The ceiling is an artefact, inconsistent with equilibrium, and set in an artificial comparative static. | answered | M10 bounds $k^*$ for every $Q$ with no limit taken. |
| R2.5 | A static one-shot model would be simpler. | answered | The game is static and played once (Appendix D). |
| R2.5 | Give the incumbent a lower cost instead of relying on concavity. | answered | M5 to M8 hold for any ordered costs, whatever their source, and concavity is replaced by burden-monotonicity (M23, M24). |
| R2.5 | Financial constraints are not crucial. | answered | No budget constraint is used beyond a challenger at or below the fee not entering (Appendix D). |
| R2.5 | Consider a reduced-form contest success function. | partly answered | The score is kept, and every result is stated in the laws $F$, $G$ and $C$ of the scores. |
| R2.5 | Why is $\mu < \frac{1}{2}$ needed? | answered | No row assumes it. The family of conjecture R1 uses $\mu \leq \frac{1}{2}$ only to be explicit. |
| R2.6 | Results should be precise propositions with proofs. | answered | Each model row states hypotheses and a conclusion and has a proof checked in Lean (Appendix A). |
| R2.6 | Errors in the text of the earlier manuscript (saving, $p_1$, marginal utility cost, income ratios, free riding, the Sutton Trust passage). | moot | The text is retired. One point carries over. A uniform rise in every wealth lowers every cost, so the count does not fall (M12). |

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